

Some pages covering:

**RELIABILITY ANALYSIS OF
NETWORKS OF CONTROL POINTS:**

**GROSS ERROR TEST
GLOBAL TEST
INTERNAL RELIABILITY
EXTERNAL RELIABILITY**

**For use in the course
TBA4245 GEODESY**

Section of Geomatics

Department of transport and building science

NTNU

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Some pages covering:

RELIABILITY ANALYSIS OF NETWORKS OF CONTROL POINTS:

- **GROSS ERROR TEST (*GROVFEILTEST*)**
- **GLOBAL TEST**
- **INTERNAL RELIABILITY (*INDRE PÅLITELIGHET*)**
- **EXTERNAL RELIABILITY (*YTRE PÅLITELIGHET*)**

For use in the course TBA4245 Geodesy

In the course TBA4245 Geodesy it is important to understand the different steps in reliability analysis and to have a basic knowledge of the theory. Understanding a printout from a program is part of the syllabus.

Contents:

A note covering analysis and reliability of a network of control points by Terje Skogseth.

Part A: The contents is copied from chapter 9 in the Norwegian textbook "Grunnleggende landmåling" (Basic land surveying) by Terje Skogseth.

Part B: The contents in part A is considerably extended in part B concerning.

- Gross error test
- Test of the coordinates and heights (Global test)
- Reliability analysis (internal and external)

Other references used:

Sverre Wisløff, Norkart AS. Lecture notes and compendium from 1998, gross error detection.

Knut Ragnar Holm, Geomatics NTNU: Power points presentation in reliability analysis.

The user manuals of the Norwegian computer programs GIS/Line and Gemini.

Standard from the Norwegian Mapping Authorities. The geodata standard. In Norwegian.

All the standards covering mapping and surveying can be found in the home pages of the Norwegian Mapping Authorities, under "Standardisering".

<http://www.statkart.no/>

See also the example exercises on It's learning covering the themes in this note.

This note is originally written in Norwegian and it was translated to English as an English version was required in autumn 2009. This is the "beta" version, and the quality control of the text and the spelling could have been better.

English version 3.11.2009

20.10.2010: Some misprints are corrected

Terje Skogseth

Note on: Analysis of networks and reliability of networks

Abstract from "Basic Land Surveying" by Terje Skogseth et al, Universitetsforlaget 1998 Chapter 9: The quality of networks (brief overview, examples). The abstract is expanded in terms of:

- Gross error detection
- Test of the coordinates of known points (Global Test)
- Reliability analysis (internal and external)

The chapter is an extension of the theme as it is processed in “Basic Land Surveying”.

In the course Geodesy, it is important to understand a printing from a reliability computation, and also have knowledge of the theory behind. The theory is normally well described in the user manuals for surveying applications (in Norway: Gemini and GIS/Line), and these manuals are also used as background material for this chapter. The chapter is not quite in depth within the individual themes, it is done in later courses.

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A. Background, about networks of control points

Chapter A has the same content as part of Chapter 9 in “Grunnleggende landmåling” (Basic Land Surveying). Headwords are: The definition of different networks, planning and construction of a GPS network. In chapter 9.6 of the textbook the new principles and mindset when it comes to control of quality in the new Norwegian *Geodata standard* (which replaces the *Norm of 1982*) is demonstrated and explained.

A.1 Methods of measurements

A.1.1 Introduction

The field work in connection with establishing new control points is often done by establishing a network of known and new control points. Condensing and maintenance of the control points is an important task.

The most commonly used methods for densifying the network of control points, are:

- Satellite measurements (GPS / GNSS)
- Measurements performed with traditional surveying equipment:
 - Triangulation and trilateration
 - Traverses
 - Single point measurements

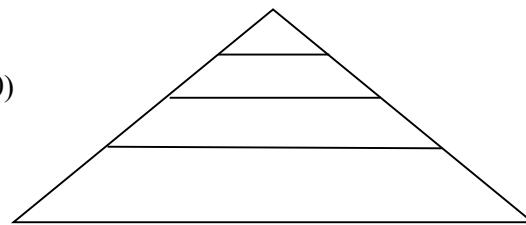
In the adjustments all the observations are used in a simultaneous computation. The adjustment is based on the principles of "least squares method", which gives the lowest standard deviation on the adjusted coordinates and heights. In this note we do not go into this general adjustment method (which is also used in computer programs).

A.2 Networks of control points

The Norwegian Mapping Authority is responsible for the fundamental national networks of control points. New and more precise measuring instruments (the use of satellite positioning and long-range electronic distance meters) have revealed a very variable quality of the coordinates and heights calculated by the older measurements, for instance the old Norwegian datum NGO1948.

The control points can be divided into several networks: Control points in coarse-meshed first order network of control points, to points for detailed measurements. An overview showing the hierarchy of the control points in Norway:

- National fundamental points (ca 100)
- National basic points “Stamnett” (ca 1000)
- The first order net of the municipalities, “Landsnett” (ca 13000)
- Control points for setting out and measuring-in “Bruksnett”



The old network in Trondheim consisted of 15 000 control points. The new official network has 500 points, and these 500 control points are part of the “Landsnett”, see above.

GPS (GNSS) is widely used today for detail measurements (setting out and measuring-in). How many detail points do we need today? We can use permanent base points where continuously measurements with GNSS receivers are carried out. Using base points and relative GNSS (GPS) measurements makes it possible to manage with just 1-5 base points to carry out the GNSS measurements.

The Norwegian Mapping Authority has carried out comprehensive measurements and computations in 1996 and 1997 of their networks, using satellite measurements (GPS) and precise

levelling (heights). The control points in the network have thus been given *final values* in the datum EUREF89, *the frame*. These points and their coordinates and heights are the *realization* of the definition of EUREF89, *the system*. The "Landsnett", see above, provides further condensing of control points and is measured primarily by GPS. The "Landsnett" will typically be the main municipal network. The next level is a detailed network called "Bruksnett" and will provide points for detailed measurements, and these points can also be measured in with other methods than GPS. It is the municipal surveying authority which has the main overview of the municipal primary and detailed network of control points.

On the homepage of The Norwegian Mapping Authority you will find the coordinates and heights of the control points in the "Stamnett" and the "Landsnett" and the heights of the national first order bench marks.

A.3 Networks. Measuring methods

Measurement of new control points is usually carried out in connection with an extension or densification of existing network, or in connection with larger projects (such as new housing development, new large bridge or a new major road construction).

- Satellite measurements have taken over more and more of the measurements of new control points. This is true not only for measurements in the main municipal network, but also for setting out or measuring-in points which earlier have been determined by traverses and single point determination by use of a total station or a theodolite.
- A total station (a precise theodolite and electronic distance measure in one unit) can also be used to determine a classic network of control points. A classic network can be a combination of triangulation, trilateration (distance measurement) and traverses. This method is now only used where there are problems getting signals from the GNSS satellites, for instance between high buildings in city or close to or inside a tunnel.

Both methods have common characteristics in terms of planning, measurement and calculation.

Satellite Measurements

In Chapter 8 in "Basic Land Surveying" different measurement methods for point determination with GPS are described. For surveying and condensing of the municipal networks, the "classic" method of building up a network of GPS baselines is most commonly used. See an example in figure A.1. *Independent* baselines must be used in planning and computation of the network.

A.4 Network. Planning of the layout of the network

The planning of a network requires some experience. Previously, we were very concerned with good cuts, while the introduction of the electronic distance meter and the satellite receiver has changed the requirements for the geometric design. The network is built up with a good judgement, where we make sure that each new point has a good geometry concerning the GPS baselines.

The new control points must be determined by more measurements than "necessary", with excess (extra) measurements. In larger new networks the excess measurements constitute 30-40% of the measurements. If there are 10 new control points to be determined (20 unknown coordinate values, x and y), it is in principle enough with 20 measurements, one for each unknown. When, for example, we have a total of 30 plane bearings and distance measurements to determine the 10 points, and then there are 10 additional measurements for control and to obtain good coordinates on the new points. It is important that each new coordinate is measured with excess measurements.

As for the highest order networks we would like to have at least 50% of excess measurements. The reason for this is:

1. We can achieve higher accuracy of the points.
2. We get a better measure of the achieved accuracy.

3. We have the ability to verify the accuracy of the known points.

Simulation of a network

It is also developed software that can help us in planning the network. If we specify an approximate location of the new points and the measurements we will perform (and the presumed accuracy of these), the simulation program can calculate the theoretical accuracy we get on the new points. This is called simulation computation. It is important that we are aware of that the known control points normally are assumed to be error-free (have standard deviation of 0) in these programs.

Design of a network

It is advisably to make the following in the design of a network:

- It is especially important to connect adjacent (neighbouring) points.
- It is important that the network will be well connected to the existing known control points. We should therefore make connections to the existing points around the network.
- Draw a connection line (polygon) between the known control points. The bulk of the network of the new control points should be within this polygon.
- We obtain a control of the existing known points when some extra measurements are carried out in such a way that the known points can be treated as unknown points (to be able to check their coordinates and heights). For example: When GPS is used, we should also measure the baselines between the known points, see figure A.1.

Building up the GPS network

When total stations are used in the measurements, the network is built up by using triangles.

A GPS network can consist of squares of independent baselines. A "square" normally provides a sufficient number of excess measurements. One of the reasons why the GPS network is still of good quality, is that a baseline has a plane bearing as one of the measurements (the other two measurements are the distance and zenith angle). The bearing, distance and zenith angle can be calculated from the coordinate differences of the base line (ΔY , ΔX , ΔZ).

Example of a GPS network

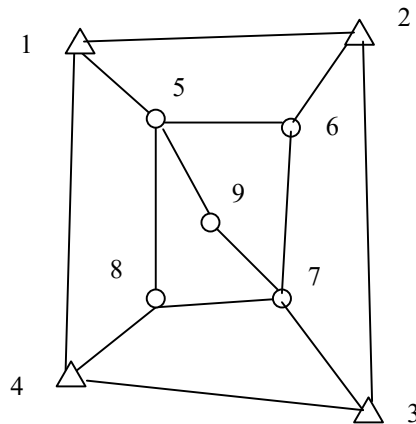


Figure A.1 GPS network

In figure A.1 the points 1, 2, 3 and 4 are known points, the other 5 are unknown points whose coordinates and heights are to be determined. A total of 14 independent baselines are measured, where 4 is between the known points to obtain a control of the coordinates and heights of the known points.

Number of measurements: 14 baselines, 3 measurements per baseline (ΔY , ΔX , ΔZ): $n = 42$

Number unknowns: 5 new points, 3 unknowns per point (Y , X , Z): $e = 15$

System unknowns: Scale, rotation, vertical deviation (east and north value): $e_s = 4$

Number of excess measurements, the degree of freedom: $f = n - e - e_s = 23$

In this example, we have approximately 100% excess measurements: 42 measurements to determine the 19 unknowns and for an inspection of the coordinates of the known points.

A.5 Methods for giving the accuracy of the control points

From the adjustment of the network we obtain:

- (1) Adjustment / estimation of the unknowns (x, y, H)
- (2) The accuracy of the unknowns in (1)

Accuracy measures:

1-dimensional: Standard deviation

2-dimensional: Error ellipse

A.5.1 The standard deviations and the error ellipse

From a compendium of Torgeir Torkildsen, here about the error ellipses:

The error ellipse has the same meaning as the standard deviation in a one-dimensional interval

The confidence coefficient is 0.39 (39%) compared to 0.68 in a one-dimensional case. The reason for a lower value: Two events in the two-dimensional interval: x and y shall both at the same time be within their one-dimensional interval, see figure A.2.

To achieve the same confidence coefficient, 0.68, the standard ellipse must have a linear enlargement of 1.5. In area this will give an enlargement of 2.3.

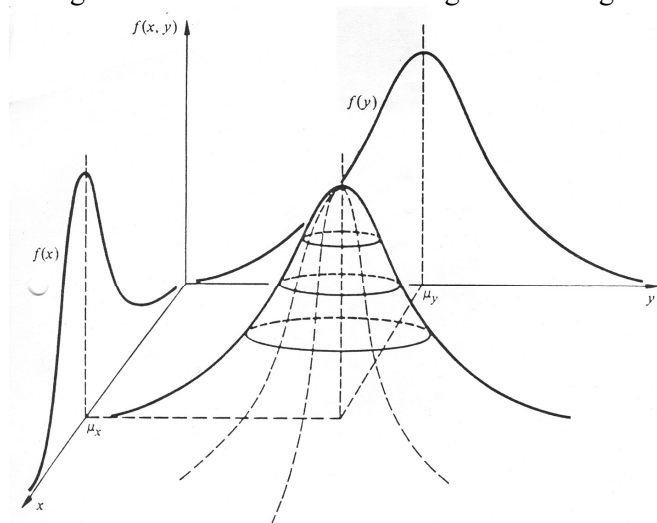


Fig. 8-6.

Figure A.2: The standard deviations and the error ellipse

Different error ellipses:

- (1) With **known standard deviation**, σ_0^2 :

Linear factor of enlargement for the standard ellipse k	Confidence coefficient α	Name
1.00	0.39	<i>Standard ellipse (Helmert ellipse)</i>
1.18	0.50	Probable error ellipse
1.31	0.57	Average error ellipse
1.52	0.68	95% confidence ellipse. Average error ellipse, Wellish
2.45	0.95	

Table A.1 Ellipses, known standard deviation

(2) For most of the adjustments the **standard deviation, σ_0^2 , is assumed to be unknown**
 The table shows the 95% confidence ellipse, ($n-e$) is the number of redundant observations:

k	The degree of freedom ($n-e$)
20.0	1
6.16	2
3.40	5
2.86	10
2.52	50
2.45	∞

Table A.2 Ellipses, assumed standard deviation

We recognize $k = 2.45$ for $n - e = \infty$ in table A.1.

Error ellipses in a network of control points

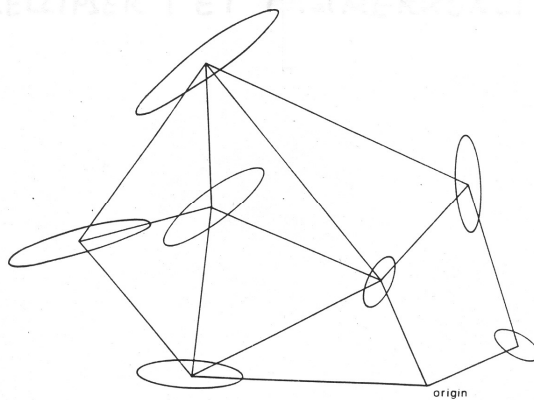


Figure A.3 Free network and error ellipses

Error ellipses: Which factors decide the size and the direction of the axis?

- Geometry, the measurements. The more measurements to/from a point the lower values of the axis of the error ellipse of the point
- The distance from the known points. The further away, the bigger values of the axis of the error ellipse of a point.
- The error ellipses are independent of the axis of the coordinate system.

Some disadvantages of using error ellipses as measures of accuracy:

- The known control points are normally considered as faultless (the axis of the error ellipse are zero). Not true, as these points have been calculated in an earlier project.
- The distance from the known points. The further away, the bigger values of the axis of the error ellipse of a point.

Conclusion: We must find a better way to determine if the network and the new control points are fulfilling the tolerances concerning the quality of the coordinates and the heights.

A.6 Error categories

The observations measured by the total station, the leveller or the GPS receiver are or could be erroneous. We divide these errors into three groups:

- Systematic errors
 - The functional model does not reflect well the physical reality
 - May be avoided with a better functional model,
 - or by compensating measuring techniques
 - Ex.: Measuring of angles in rounds (theodolite), levelling techniques. Calibrating or checking the instrument.
- Gross errors
 - Mistake, erroneous measuring, identification error
 - Software errors (!)
- Random errors
 - Natural variations in observed quantities
 - Must be taken into consideration in the stochastic model, and in estimation

Avoid gross errors:

- Gross errors can only be avoided by ensuring they don't come into existence
- Good measuring procedures
- They are detected and removed, but smaller gross errors nevertheless exist
- Good methods/tools for error detection

Example: A distance is measured 10 times

Observ. no	Measured value (m)	Corrected observation, l	$v = x - l$ (mm)	v^2
1	998,843	998,852	+6,7	44,89
2	998,850	998,859	-0,3	0,09
3	988,847	998,856	+2,7	7,29
4	998,846	998,855	+3,7	13,69
5	998,844	998,853	+5,7	32,49
6	998,861	998,870	-11,3	127,69
7	998,884	(not used)	(not used)	-
8	998,853	998,862	-3,3	10,89
9	998,855	998,864	-5,3	28,09
10	998,848	998,857	+1,7	2,89
		$x = 998,8587$	$\sum v = 0,3$	$\sum v^2 = 268,01$

Table A.3 Observations and corrections

Gross errors:

- The value of no 3 in the “Measured” column differs with 10 m from the other 9 observations. This observation is corrected by 10 m and thus the gross error is eliminated.
- No 7 is assumed to be a gross error as the value differs too much from the other 9 observations. A statistical testing will show if this assumption is correct.

Systematic errors:

The distance measure has been calibrated (checked) and the calibration shows that this instrument has a constant error of 0.009 m. The corrected values are shown in the “Corrected observation” column.

Most probably value of the distance:

All the observations are assumed to have the same accuracy, the same weight. Then the arithmetical mean is the most probably value of the distance.

B. Analysis and Reliability

Estimation by the least squares method assumes that the observations and the control points are free of gross errors. A gross error in the calculation will be spread over the network and it can be difficult to detect the gross errors from the estimated residual errors, v , after the estimation. To improve the understanding of how serious the error affects the calculation results, it is recommended to perform some calculations in which we deliberately enter a gross error in an observation.

In the Norwegian Geodata standard, the following procedure is described for quality assurance:

1. Gross error detection: Examine the observations for gross errors. Detect and eliminate.
2. Global test: Check if the coordinates and heights of the given control points are in order
3. Internal reliability: Make an internal reliability, a test of how well the observations verify each other
4. External reliability: Make an external reliability, a deformation analysis of the network

Below is a brief description of the procedure for this analysis and the reliability test.

B.1 Gross error detection

Gross error on the observations

It can be difficult to locate gross errors. If there are measured with many excess measurements, the observations can verify each other. For the GPS network in Figure A.1, a test can be performed to check if an observation fits in with the others or not. This is done by:

1. calculating the net as a free network, where only a minimum number of known points are kept fixed (for example, the control point 1 in figure A.1) and the other known control points are considered as unknowns
2. calculating the free network without one of the observations (for example, a measured distance)
3. calculating the estimated value of the actual observation from the adjusted coordinates obtained from the calculation of the free network in point 2 above
4. comparing the original value of the observation with the estimated value of the observation from 3. If the difference is too large, it is a gross error in the actual observation
5. The same calculations are done for all the observations in the grid (n observations give n calculations in points 1-4 above)

The gross error is calculated together with the other unknowns in the estimation. Only 1 gross error is calculated in each adjustment (computation), so it must be performed as many computations as it is observations.

By using a *free network* in the calculations, any errors in the coordinates and heights of the known control points will not destroy this test. A free network means that we keep one point fixed by GPS measurements and a point and a direction fixed by a network of classical measurements (total station).

We must not exclude observations before we are fairly certain that we have located the error. It is therefore recommended to seek for a logical explanation, or eventually make a new measurement of the actual observable.

We should not discard/reject observations when the gross error test on the observations is based on a constrained adjustment or a constrained network. Errors on the coordinates of the known points that are kept fixed, could damage the test.

By a gross error test, we can locate one or more gross errors in the observations, with a certain probability. After observations with proven gross errors are removed or corrected, we have a set of observations that apparently is in order.

An important result, in addition to the detection of the gross errors, is that gross error test gives us an estimate on how large the maximum values of the remaining gross errors are (statistical values).

B.1.1 The definition of the remaining gross errors

Errors that are not big enough to be determined with sufficient certainty to be a gross error of an observation can be found by the test.

The size of the largest possible gross error on an actual observation (LGE)

By knowing the size of the largest possible gross error which can not be detected, we can proceed with this maximum remaining gross error and see how it propagates for example to the adjusted coordinates and heights of the unknown points (see the external reliability).

NB: Please note that the largest remaining gross error on an observation, is a *statistical error* determined by the geometry, the observations and the accuracy measures. If the observation network is tight or dense around an actual observation, then the other observations are controlling the actual observation well. In this case the statistically largest remaining gross error of the actual observation is small.

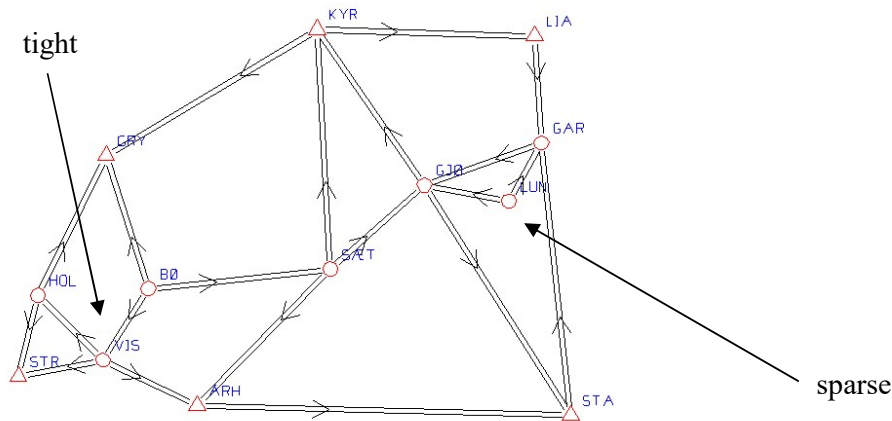


Figure B.1: A network of GPS control points

We get the opposite conclusion with a sparse network: A sparse network (few observations) gives larger value of the remaining gross error.

B.1.2 Test method, gross error detection

We use a multiple student-t-test where the estimated gross error is tested against the standard deviation of the same gross error.

The testing of a gross error is to reject H_0 (no gross error) if the estimated value on the left side of the test, is greater than the table value in the following Student t-test:

$$\frac{\hat{\nabla}_i}{\hat{\sigma}_{\nabla i}} > T_{\alpha/2, f}$$

$\hat{\nabla}$ is the calculated / estimated (gross) error,

$\hat{\sigma}_{\nabla}$ is the standard deviation on $\hat{\nabla}$, based on the estimated standard deviation of the observation of unit weight

T is the table value of the Student-t distribution with given the confidence level (test size) and the degree of freedom.

α is the confidence level (or test size) of an observation, for example, 0.1%

f is the number of excess measurements (degree of freedom)

A *confidence level* (or *test size*) of 0.1% on a gross error test for 1 observation, may seem small. It is a desired value of the total confidence level for a test of all observations (multiple t-test), which must be taken into account. If we assume that the estimated gross errors are independent of each other, the total confidence level is equal:

$$\alpha_{TOT} = 1 - (1 - \alpha)^n \quad , \text{ where } n \text{ is the number of observations}$$

- (1) Is the number of observations greater than 50, we use $\alpha = 0.1\%$ for a single test.
- (2) If $n = 50$ and $\alpha = 0.1\% = 0.001$, then the total confidence level is $\alpha_{TOT} = 0.05 = 5\%$, which is the most common numeric value of the total confidence level.
- (3) With fewer than 50 observations, we should calculate a value of α (for 1 observation) with $\alpha_{TOT} = 5\%$ as the starting point. Thus, we can be ensured that α_{TOT} has the commonly used value of 5% in the test.

With 20 observations, we obtain the confidence level of 1 observation by inverting the formula of α_{TOT} :

$$\alpha = 1 - (1 - \alpha_{TOT})^{1/n} \quad , \text{ where } \alpha = 0.0026 = 0.26\% \text{ when } \alpha_{TOT} = 5\% \text{ and } n = 20$$

Non-testable observations

Observations that cannot be controlled by other observations are called non-testable observations. Examples of such non-testable observations are:

- Polar measurements, both with total station and GPS.
- Orientation directions in known point to other known point by classical measurements with a total station.

The test *internal reliability* (see later) computes a measure of the degree of controllability of an observation, *redundancy*. The redundancy is normally between 0 and 1. If the redundancy is less than 0.005, it is considered as not testable.

Baarda's data snooping

Baarda's method to detect gross errors, data snooping, is not covered in this note (not curriculum). A comparison with the multiple-student-t-test gives some arguments for why we use a t-test instead of data snooping.

Multiple student-t-test compared with Baarda's data snooping:

- *Sharper* than data snooping (better to detect errors)
- It is based on the *estimated* value of standard deviation
- It is also valid for *correlated* observations (e.g. decomposed GPS baselines).

The *minimum detectable fault (MDF)* is associated with data snooping as a method to determine gross errors.

B.1.3 Statistical testing, gross error detection

Earlier criteria for gross error:

"Residual > 3 times standard error means gross error"

Corresponds to 0.27% gross errors with Normal distribution

From experience: 1-5% gross errors in photogrammetry / land surveying

Residuals (adjustment corrections) – not the best test quantity

Statistical testing gives more reliable result. Must use the correct test quantity

By carrying out a gross error test, we can locate one or several gross errors on the observations with a certain possibility.

A gross error test gives us also an estimate on how large the remaining gross errors are

Relevant questions

- **Should one of two models be rejected?**
Example of model alternatives:
“The residual is random” vs. “The residual is a gross error”
- **How large error is needed to reject a model?**
Non-centricity parameter δ_0 . Minimum deviation can be much larger than the test threshold
- **How large is a gross error in a suspicious observation?**
Redundancy number r_{ii} . Generally, only a small part appears in the residual of the observation
- **What is the smallest gross error that can be detected?**
Controllability $\nabla_0 l_i$. Normally much larger than three times standard deviation
- **How will one gross error influence on the result?**
- Influence $\nabla_i x$ – Decisive for possible elimination of the observation or parameter
- What is maximum influence on the result from a not detected gross error?
- Sensitivity $\nabla_0 f$ – May be used to assess the design

B.1.4 Statistical testing

Gross error detection using a multiple t-test

Test:

H_0 = no gross error ($\nabla = 0$)

H_a = a gross error exists ($\nabla \neq 0$)

Error type 1: α = the probability to reject H_0 when H_0 is correct

Error type 2: $1 - \beta$ = probability of accepting H_0 when H_0 is false

The probability $(1 - \beta_0)$ (β_0 = test strength) is the probability of not detecting any gross error even if it actually exists. The probability of ignoring a gross error, error type 2

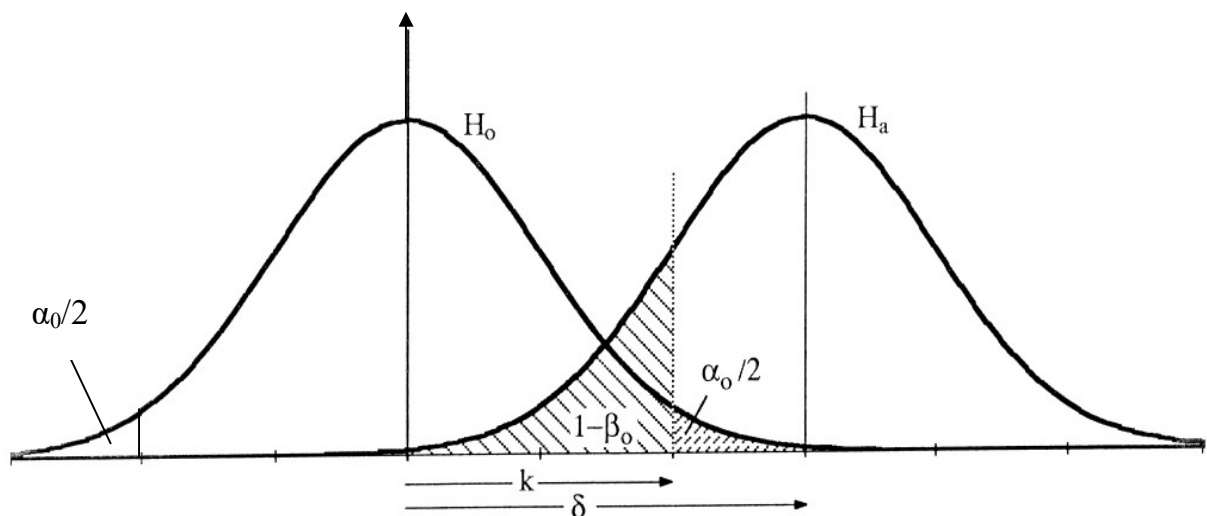


Figure B.2: Two tailed test. The density function of observations without a gross errors (left) and of the observations with gross errors (right)

α_0 : Confidence level (or test size), the probability of type 1 error: Reject an observation wrongly, believing it is an observation with a gross error

β_0 : Test power. Which limit must a gross error cross to be detected with a security of β_0 ?

$(1 - \beta_0)$: The probability of type 2 error. Accept the observation wrongly, believing it is an observation without a gross error when it has a gross error. The probability $(1 - \beta_0)$ (β_0 = test power) is the probability of not detect any gross error even if it actually exists. The probability of ignoring a gross error, error type 2

δ_0 : Threshold (limit) for residual that is rejected with chosen test power/significance level

k : Threshold value of the test.

More precise: The distribution of the alternative hypothesis is a non-central student's-T-distribution (which has a more complicated density function).

Definition of the remaining gross error:

A remaining gross error is an error that is not large enough to be characterized with sufficient certainty as a gross error of an observation.

We know the size of the largest possible gross error that cannot be discovered:

How will this largest possible gross error propagate to errors of the coordinates and heights of the unknown points (see *external reliability*)?

NB: Notice that the largest possible gross error of an observation is a statistical value which is dependent on the geometry, the observations and the measures of the accuracy.

Maximum relative gross error (MRG)

MRG is defined as the largest remaining gross error in multiples of the standard deviation of the observation:

$$\frac{\nabla_{0i}}{\hat{\sigma}_i}$$

If the maximum relative gross error has a value of 3, we can argue that the actual observation does not contain a gross error greater than three times the observation own standard deviation, with the selected significance level.

Generally: We can have a remaining gross errors that may be about 7 to 8 times the standard deviation without being detected by the statistical testing, see for instance example exercise no. 9 (gross error).

Examples from a statistical table

Corresponding values of k , α_0 , β_0 , δ_0 . Normalized errors.

k	α_0	β_0	δ_0
1.280	0.200	0.500	1.280
1.280	0.200	0.800	2.120
1.280	0.200	0.900	2.560
1.280	0.200	0.950	2.920
1.280	0.200	0.990	3.610

k	α_0	β_0	δ_0
2.500	0.012	0.500	2.500
2.500	0.012	0.800	3.340
2.500	0.012	0.900	3.780
2.500	0.012	0.930	4.000
2.500	0.012	0.950	4.140

1.640	0.100	0.500	1.640
1.640	0.100	0.800	2.480
1.640	0.100	0.900	2.920
1.640	0.100	0.950	3.280
1.640	0.100	0.990	3.970
1.640	0.100	0.991	4.000
1.960	0.050	0.500	1.960
1.960	0.050	0.800	2.800
1.960	0.050	0.900	3.240
1.960	0.050	0.950	3.600
1.960	0.050	0.980	4.000
1.960	0.050	0.990	4.290
2.000	0.046	0.500	2.000
2.000	0.046	0.800	2.840
2.000	0.046	0.900	3.280
2.000	0.046	0.950	3.640
2.000	0.046	0.980	4.000
2.000	0.046	0.990	4.330

2.500	0.012	0.990	4.830
2.580	0.010	0.500	2.580
2.580	0.010	0.800	3.420
2.580	0.010	0.900	3.860
2.580	0.010	0.920	4.000
2.580	0.010	0.950	4.220
2.580	0.010	0.990	4.910
3.000	0.003	0.500	3.000
3.000	0.003	0.800	3.840
3.000	0.003	0.840	4.000
3.000	0.003	0.900	4.280
3.000	0.003	0.950	4.640
3.000	0.003	0.980	5.000
3.000	0.003	0.990	5.330
3.290	0.001	0.500	3.290
3.290	0.001	0.760	4.000
3.290	0.001	0.800	4.130
3.290	0.001	0.900	4.570
3.290	0.001	0.950	4.930
3.290	0.001	0.960	5.000
3.290	0.001	0.990	5.620

Table B.1 Corresponding values of k , α_0 , β_0 , δ_0

Three situations, see also the figure B.3 below:

(1) Common choices: $\alpha = 0.001$ (0.1%) and $\beta = 0.80$ (80%). By choosing these values we obtain a threshold value of $k = 3.29$.

(2) Choices if we can control the observations easily: Increase α and β , making it more easier to make a type 1 error, reject Ok observations, and making it more difficult to accept observations with gross errors (type 2 errors). In the table: $\alpha = 0.050$ (5.0%) and $\beta = 0.99$ (99%). Observe that the value of k is 1.96 in this case, compared to 3.29 above.

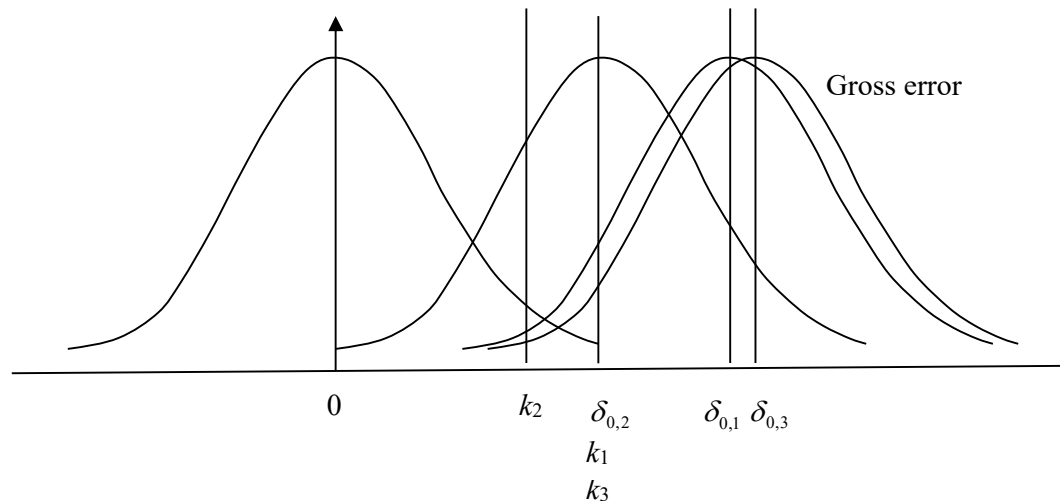


Figure B.3: 3 density function of the gross error distribution.

(3) *Choices if we do not want to reject good observations:* Decrease β , making it easier to accept observations with gross errors (type 2 errors). In the table: $\alpha = 0.001$ (0.1%) and $\beta = 0,500$ (50%). Observe that the value of k is 3.29 in this case, the same as in case (1) above.

B.1.5 The effect of a gross error on the estimated unknowns

Gross error in the observations: $\nabla \mathbf{F}$

Gross error on the unknowns: $\nabla \mathbf{X}$

The adjustment by the parameters:

Observation equations give the corrections on the observations:

$$\mathbf{V} = \mathbf{A}\mathbf{X} - \mathbf{F}$$

The normal equations give the adjusted values of the unknowns:

$$\mathbf{X} = (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} \cdot \mathbf{A}^T \mathbf{P} \mathbf{F} = \mathbf{B} \mathbf{F}$$

The connection between $\nabla \mathbf{F}$ and $\nabla \mathbf{X}$:

$$\mathbf{X} + \nabla \mathbf{X} = \mathbf{B} \cdot (\mathbf{F} + \nabla \mathbf{F}) \quad \text{and we obtain:} \quad \nabla \mathbf{X} = \mathbf{B} \cdot \nabla \mathbf{F}$$

The influence depends on the design matrix, $\mathbf{A}$, and the weight matrix, $\mathbf{P}$. A gross error on an observation is spreading to all the adjusted coordinates.

A figure showing this effect:

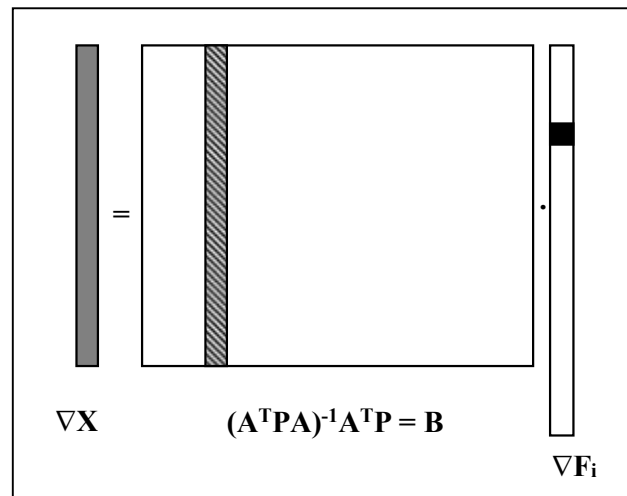


Figure B.4 The effect of a gross error on the estimated unknowns

One estimated correction in $\nabla \mathbf{X}$ is influenced by possible gross errors in all observations. See figure B.5.

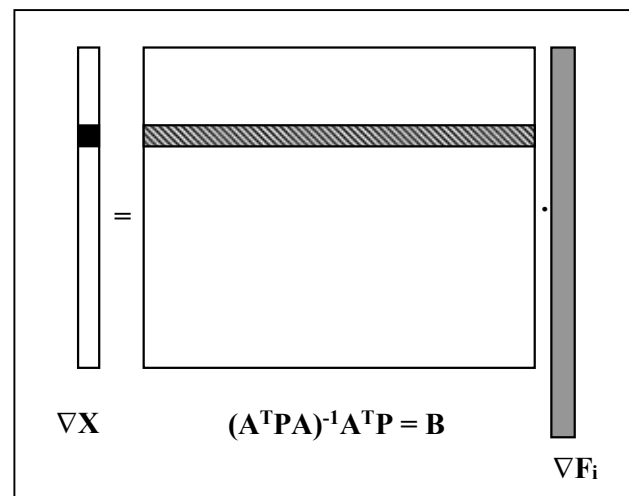


Figure B.5 The effect of a gross error on the estimated unknowns

B.1.6 The effect of a gross error on the estimated corrections of the observations

$$\text{The normal equations: } \mathbf{X} = \mathbf{Q}_{xx} \cdot \mathbf{A}^T \mathbf{P} \mathbf{F} \quad \longrightarrow \quad \mathbf{A} \mathbf{X} = \mathbf{A} \mathbf{Q}_{xx} \mathbf{A}^T \mathbf{P} \mathbf{F}$$

$$\text{The co-factor matrix: } \mathbf{Q}_{vv} = \mathbf{P}^{-1} - \mathbf{A} \mathbf{Q}_{xx} \mathbf{A}^T \quad \longrightarrow \quad \mathbf{A} \mathbf{Q}_{xx} \mathbf{A}^T = \mathbf{P}^{-1} - \mathbf{Q}_{vv}$$

$$\text{The observation equations: } \mathbf{V} = \mathbf{A} \mathbf{X} - \mathbf{F} \quad \longrightarrow \quad \mathbf{A} \mathbf{X} = \mathbf{V} + \mathbf{F}$$

$$\Rightarrow \quad \mathbf{V} = -\mathbf{Q}_{vv} \mathbf{P} \mathbf{F}$$

Assume gross errors on the observations. The effect on the results:

R: The redundancy matrix. For details about the redundancy matrix, see *internal reliability* later.

$$\nabla \mathbf{V} = -\mathbf{Q}_{vv} \cdot \mathbf{P}_{LL} \cdot \nabla \mathbf{F} = -\mathbf{R} \cdot \nabla \mathbf{F}$$

Gross error on the observations, spreading to 1 correction of the observation:

- Gross error in one observation will have an influence on all estimated corrections
- Not necessarily largest influence on "its own" correction, v_i , of the observations

1 gross error, spreading to the correction of the observations: A gross error on an observation is spreading to all the adjusted corrections on the observations. Figure B.6 shows this effect:

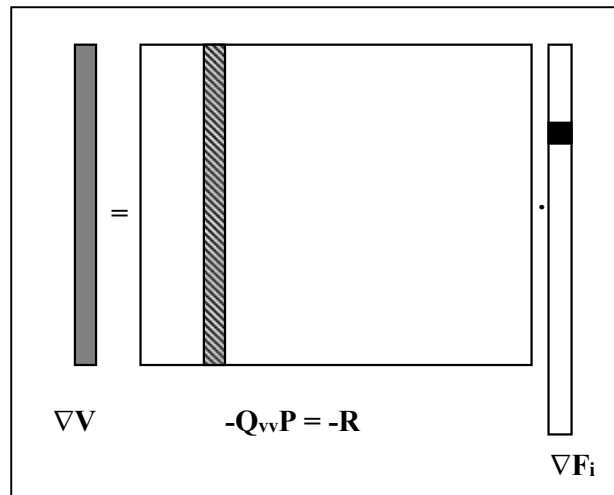


Figure B.5 The effect of a gross error on the estimated unknowns

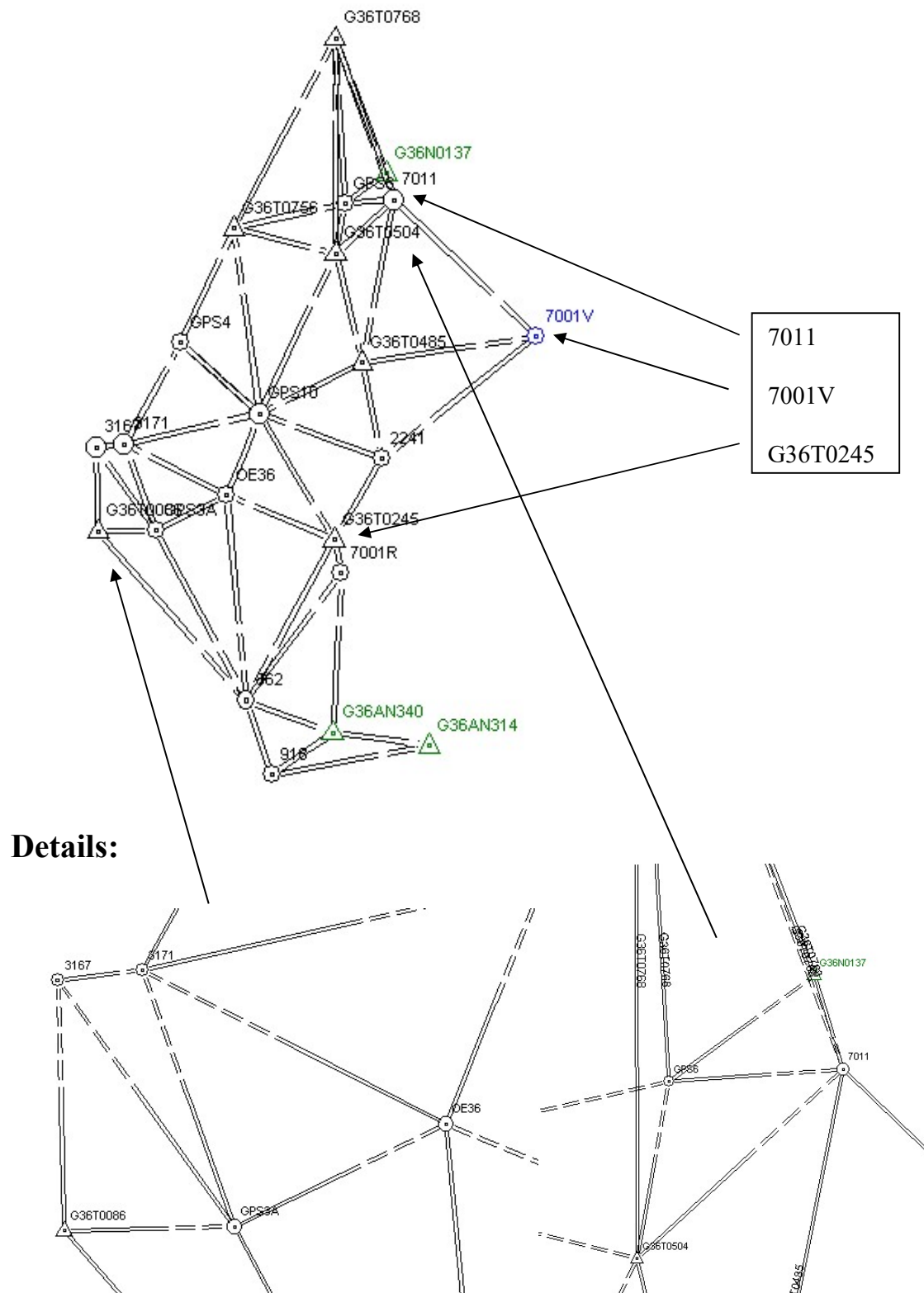
B.1.7 Influence from gross errors – summary

- Influence on estimated unknowns $\nabla \hat{\mathbf{x}} = (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P} \nabla \mathbf{F} = \mathbf{Q}_{\hat{x}\hat{x}} \mathbf{A}^T \mathbf{P} \nabla \mathbf{F}$
- Influence on estimated observation corrections $\nabla \hat{\mathbf{v}} = -\mathbf{Q}_{\hat{v}\hat{v}} \mathbf{P} \nabla \mathbf{F} = -\mathbf{R} \nabla \mathbf{F}$
- Influence on estimated observation values $\nabla \hat{\mathbf{F}} = \mathbf{A} (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P} \nabla \mathbf{F} = \mathbf{Q}_{\hat{f}\hat{f}} \mathbf{P} \nabla \mathbf{F} = \mathbf{U} \nabla \mathbf{F}$
- Redundancy matrix $\mathbf{R} = \mathbf{I} - \mathbf{A} \mathbf{Q}_{\hat{x}\hat{x}} \mathbf{A}^T \mathbf{P} = \mathbf{Q}_{\hat{v}\hat{v}} \mathbf{P}$
- Influence matrix. Influence from gross errors on estimated observations, and thus, on other parameters $\mathbf{U} = \mathbf{I} - \mathbf{R} = \mathbf{A} (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P} = \mathbf{Q}_{\hat{f}\hat{f}} \mathbf{P}$

Basis for theory of redundancy and gross error detectability. Depending on only $\mathbf{A}$ and $\mathbf{P}$

B.1.8 Example of gross error detection

GPS-network: MORY-96 (modified). Data from a Norkart-example (the land surveying program GISLine)



(1) Computation no 1. DOCUMENTATION ADJUSTMENTS

TEST OF THE OBSERVATIONS 3-D ESTIMATION. FREE NET
 GROSS ERROR TEST OF THE OBSERVATIONS - MULTIPLE T-TEST

From	To	Residual	Est.gross error	Test value

7011	1 7001V	R*	0.00167	-0.00135
7011	1 7001V	D*	0.192	-0.300
7011	1 7001V	dH*	0.026	0.068
7011	1 G36T0485	R	0.00006	-0.00006
7011	1 G36T0485	D	-0.036	0.064
7011	1 G36T0485	dH	0.009	-0.029
G36T0485	1 2241	R	0.00033	-0.00133
G36T0485	1 2241	D	0.010	-0.019
G36T0485	1 2241	dH	-0.009	-0.028
G36T0485	1 7001V	R	-0.00588	0.00984
G36T0485	1 7001V	D	0.011	0.050
G36T0485	1 7001V	dH	0.019	-0.222
G36T0485	1 GPS10	R	-0.00001	0.00152
G36T0485	1 GPS10	D	-0.065	0.100
G36T0485	1 GPS10	dH	0.007	0.102

Tabulated value =3.41 (Student-t, f=86, alfa =0.0005)

SUMMARY:

Category	Number of obs.	Accumulated (%)

Test/Table < 1.0	147	98.00
1.0 < Test/Table < 2.0	2	99.33
2.0 < Test/Table < 3.0	0	99.33
3.0 < Test/Table	1	100.00
Not controlable	0	100.00

STATISTIKK

The number of observations, directions: 50
 The number of observations, distances: 50
 The number of observations, height differences: 50
 The number of observations: 150
 The number of unknown coordinates: 44
 The number of unknown heights: 22
 The number of additional unknowns: 4
 The number of unknowns: 70
 The degree of freedom: 87

Comments:

*: Observations: R = plane bearing D = Distance dH = height difference

The observations with detected gross errors are marked with asterisk, *
 Only 1 gross error (the biggest, ***) is dealt with before the next computation. The *** gross error is corrected (not removed) before comp no 2 on the next page. Just 1 at a time, because a bog gross error can corrupt the neighbouring observations. In this case the observation 7011 to 7001V (one *), which has point 7001V in common with the *** gross error.

(2) Computation no 2. DOCUMENTATION ADJUSTMENTS

GROSS ERROR TEST OF THE OBSERVATIONS - MULTIPLE T-TEST

From	To		Residual	Est.gross error	Test value	

G36T0485	1 2241	R	0.00024	-0.00120	4.28	*
G36T0485	1 GPS10	R	0.00020	0.00129	3.89	*
G36T0485	1 GPS10	D	-0.069	0.101	36.81	***
G36T0504	1 G36T0485	R	-0.00034	0.00091	4.17	*
Tabulated value =3.41 (Student-t, f=86, alfa =0.0005)						

Summary:

Category	Number of obs.	Accumulated (%)

Test/Table < 1.0	0	0.00
1.0 < Test/Table < 2.0	3	75.00
2.0 < Test/Table < 3.0	0	75.00
3.0 < Test/Table	1	100.00
Un-controllable	0	100.00

Comments: The *** gross error on the previous page is corrected (not removed) in comp no 2. The * gross error on observation G36T0485-GPS10 (distance) in comp 1 is a *** gross error in comp 2. The * gross error on observation 7011-7001V (distance) in comp 1 is no longer a gross error in comp no 2. Two new * gross errors in comp no 2, both observations to/from point G36T0485. NB: The degree of freedom is $86 = 87 - 1$. There is 1 additional unknown (the gross error in the actual adjustment, therefore 1 less than the f in the summary on page 22).

(3) Computation no 3. DOCUMENTATION ADJUSTMENTS

GROSS ERROR TEST OF THE OBSERVATIONS - MULTIPLE T-TEST

From	To		Residual	Est.gross error	Test value	

3171	1 GPS4	dH	0.045	-0.071	3.97	*

Tabulated value =3.41 (Student-t, f=86, alfa = 0.0005)

SUMMARY:

Category	Number of obs.	Accumulated (%)

Test/Table < 1.0	0	0.00
1.0 < Test/Table < 2.0	1	100.00
2.0 < Test/Table < 3.0	0	100.00
3.0 < Test/Table	0	100.00
Un-controllable	0	100.00

Comments: The *** gross error on the previous page is corrected (not removed) in comp no 3. There is still 1 gross error left, we should have carried out another computation, where the * observation is corrected or removed. The excuse is that the test value of 3.97 is close to the tabulated value of 3.41, but....

B.2 Global testing

We want to detect errors on known coordinates and heights (global test).

After a gross error test is performed, the observations shall not contain any gross errors. If we have measured in such a way that the coordinates and heights of the known points can be controlled, we can perform a test on the known coordinates and heights of these points.

The test makes a comparison of $\sum pvv = \mathbf{V}^T \mathbf{P} \mathbf{V}$ of a *constrained adjustment* with $\sum pvv$ of a *free adjustment*.

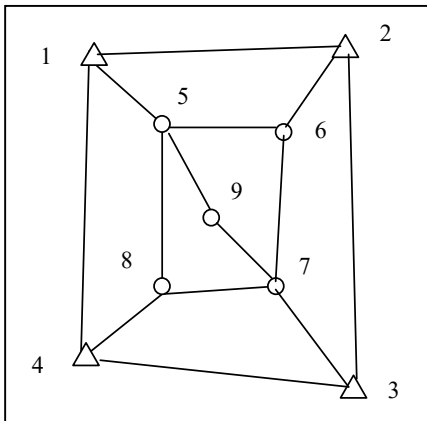
A *constrained adjustment* is an adjustment where more than 1 control point is fixed in the adjustment (of a GPS network), a constrained network.

In the constrained adjustment it is possible and also recommended to introduce some additional unknowns in the adjustment, to solve the problem with different datum of the measurements (GPS) and the national datum:

- Scale (differences in scale between the WGS84 GPS datum and the national datum, for instance EUREF89)
- Rotation (differences in rotation between the WGS84 GPS datum and the national datum, for instance EUREF89)
- Deviation of the vertical (difference between GPS measured ellipsoidal heights and orthometrical heights with the geoid as the reference plane). 2 unknowns, the north and the east component of the deviation of the vertical.

A *free adjustment* is an adjustment where only 1 point is kept fixed in a GPS network (free network). In a free adjustment any constraint caused by error(s) on the coordinate(s) and/or the height(s) make no effect, as only one control point is held fixed in a GPS measured network.

B.2.1 The method of global testing



1. Carry out an adjustment with all the known control points fixed. For instance, point 1, 2, 3 and 4 on the figure. Constrained network.
2. Carry out an adjustment with just 1 known control point fixed. For instance, point 1 on the figure. Free network.
3. Compare the constrained adjustment in 1 with the free adjustment in 2, by doing a Fisher test, see the next chapter.
4. If the test fails, there are constraints: Find the control point with erroneous values and treat this point as an unknown in the next adjustment or test.
5. Continue with points 1 – 4 above until all erroneous control points are detected by the Fisher test.

Figure B.7 GPS network

A global test can be performed on the coordinates and heights individually, or by a combined three-dimensional test.

B.2.2 Hypothesis test

We can choose between two tests, depending on which standard deviation we are using:

- (1): A *Fisher test* must be chosen when estimated values of the standard deviations are used.
- (2): A *Chi-square test* must be chosen if we use assumed value of the standard deviation.

(1): A Fisher test must be chosen when estimated values of the standard deviations are used. The Fisher test is most commonly in use as the number of excess measurements (degree of freedom) usually are so large that estimated values are better to use than assumed values of the standard deviation. The Fisher test is a hypothesis test on the two $\sum pvv$ from the free and the constrained adjustment.

We reject H_0 (the coordinates/heights of the known points are OK) if the estimated value on the left-hand side of the test is greater than the table value on the right-hand side in the following Fisher-test:

$$\frac{(\sum pvv_T - \sum pvv_F) / (f_T - f_F)}{\sum pvv_F / f_F} \geq F_{\alpha, (f_T - f_F), f_F} \quad (B.1)$$

In equation (B.1) is the index T for constrained network and the index F for free network, f is the number of excess measurements (degree of freedom), normal test value of α is 5%.

It can be shown that: $(\sum pvv_F - \sum pvv_T) / (f_T - f_F)$ and $\sum pvv_F / f_F$ are independent and χ^2 -distributed variables. Therefore the Fisher test in (B.1) is valid.

(2): A *Chi-square test* must be chosen if we use assumed value of the standard deviation from the adjustment of the free network, instead of the estimated value of the standard deviation.

We reject H_0 (the coordinates/heights of the known points are OK) if the estimated value on the left-hand side of the test, is greater than the table value on the right-hand side in the following Chi-square test

$$z = \frac{\sum pvv_T - \sum pvv_F}{\sigma_0^2} \geq \chi^2_{(f_T - f_F), \alpha} \quad (B.2)$$

B.2.3 Location of points with erroneous coordinates / heights

If the test in equation (B.1) or (B.2) shows that the coordinates/heights of the known points are not OK, we must start the search for the control point(s) that give(s) constraints in the network.

Criteria to find the control point that most probably is corrupting the result of the test:

1. The known control point that improves (reduces the numerical value) of $\sum pvv_T$ for the constrained network the most, is treated as an unknown point further on in the global test. For instance control point 2 in figure B.7.
2. Then, a new global test is performed with a new unknown point (the control point with erroneous values from point 1) in addition to those control point that already are considered as unknown control points. In figure B.7 we then have the following unknown control points: 5, 6, 7, 8, 9 and 2.
3. If the test still fails (we still have errors on the known coordinate/heights): We must carry out further testing according to point 1 and 2 above. A new known control point is found to have erroneous coordinate values and is considered as an unknown point in the next round of the global test. We stop the global test when the H_0 hypothesis is OK (no errors on the remaining known control points).
4. Over compensation test: A test showing if we have found the erroneous control points, even of the test shows that the H_0 hypothesis is OK. Another known control point is treated as unknown by using the method described above, the point which reduces the numerical value of $\sum pvv_T$ most. Then, one of the known control points which is considered as unknown is again considered as known. Here we choose the point that will increase the numerical value of $\sum pvv_T$ less. Normally it is the same control point that is released and then fixed again, but

if there are some known control points which are equally in quality, then the over compensation method could give a different result.

To be able to carry out a hypothesis test, it is assumed that the actual network has excess measurements (degree of freedom), even when one or more known control points are treated as unknown points. In deformed networks we can end up with a free network, there are no more known control points to treat as unknown points.

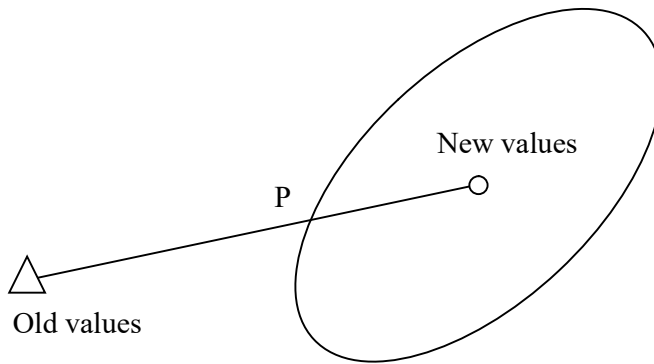
After the global test is performed, we have an overview of control points that can have erroneous coordinates and/or heights.

The next step in the control of the known coordinates and heights, is to make a closer examination of these points, see the next chapter.

B.2.4 Test of the changes in coordinates and / or heights

We should not simply dismiss the coordinate values on a known control point, which has been treated as an unknown point in the global test. It must be performed a test on whether the new coordinates / heights are significantly different from the existing values.

In addition, there may also be a political consideration whether it should be made (*small*) changes in the existing coordinates and heights. A change may have consequences for other control points and boundary marks based on the existing values of the actual control point.



The figure shows the problem of control of coordinates. The results of the adjustment of the network also give us the accuracy measures to be able to calculate a confidence ellipse (error ellipse) around the new position of the point.

For the control of the height, we are able to create a confidence interval around the new height value, equivalent to the figure.

Figure B.8 Locations of new and old positions of a control point

The point of interest on the figure is point P which is located on the intersection between the ellipse and the straight line between old and new position:

- If P is located closer to the old position (triangle) than the new location, the old value is retained.
- If P is located closer to the new location than the old position (triangle), are the new values significantly different from the old values.

It is the direction between the old and new position that is interesting:

Around the new position of P a confidence interval along the line between new and old position is calculated. A student-t-distribution is used as we use estimated value of the distance S between the old and the new position and estimated value of the standard deviation of the distance σ_s . D is half the confidence interval:

$$D = \hat{\sigma}_0^2 \cdot \sqrt{\hat{q}_{ss}} \cdot T_{\%2, f}$$

$\hat{\sigma}_0^2$: Estimated standard deviation of unit weight

$\hat{q}_{ss}$: Co-factor coefficient of the distance between the old and new position

$T_{\alpha/2, f}$: Table value with f degree of freedom and α as the confidence level

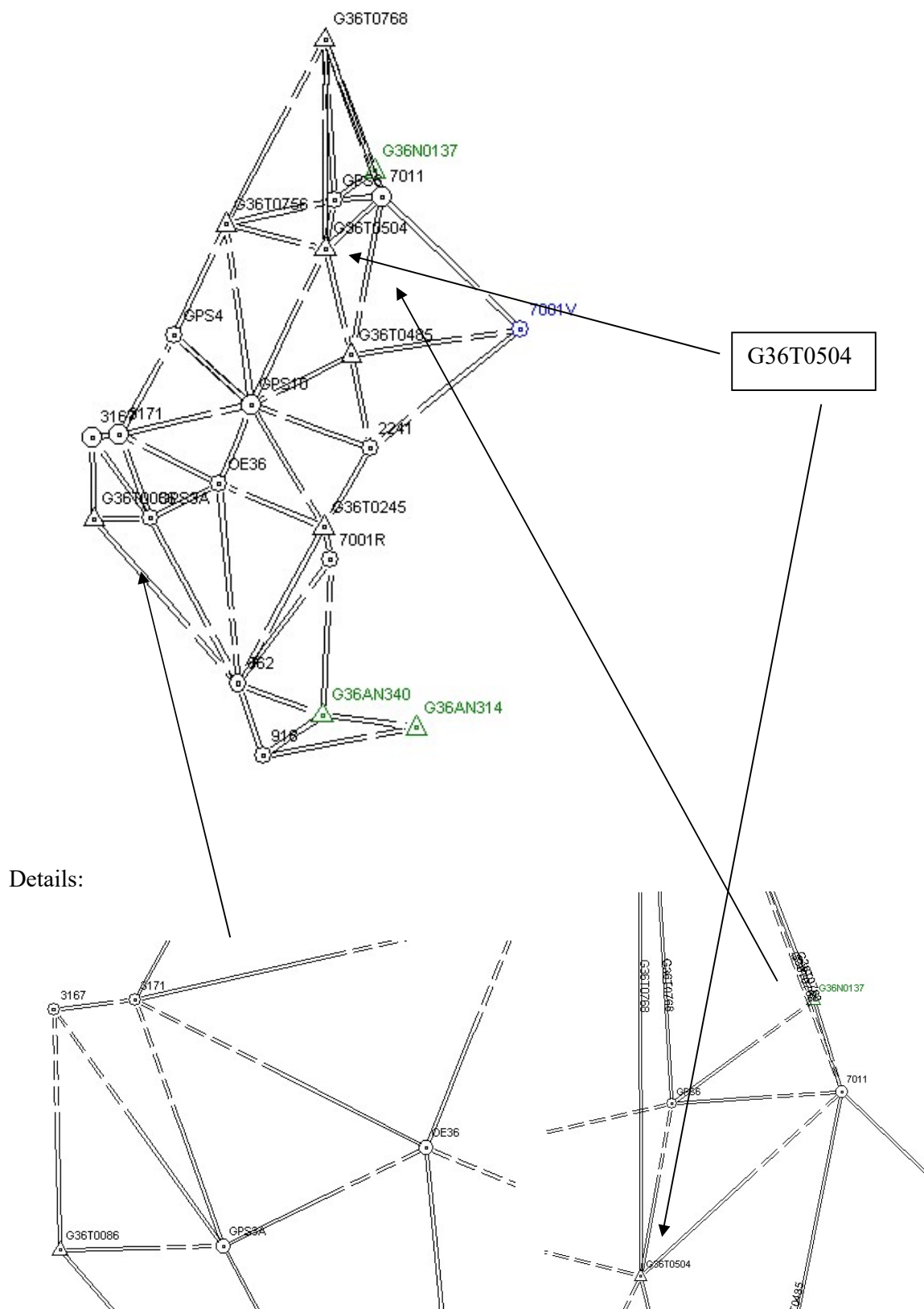
α : The test size or the confidence level. Common choice is 5%

The threshold value is equal to the limit of the confidence interval in the direction where the point is deformed (the line between old and new position).

On the figure, P is closest to the new location: a significant change in position is proven.

B.2.5 Example: Global testing

Same project as the one in gross error detection, see chapter B.1.8. GPS-network: MORY-96 (modified). Data from a Norkart-example (the land surveying program GISLine)



DOCUMENTATION ADJUSTMENTS. The network named Mory

(1) TEST OF COORDINATES AND HEIGHTS

DATUM: NGO1948 - AXIS 3

ADJUSTMENT IN 3-D

KNOWN COORDINATES [meter]

POINT	X	Y	H
-----	-----	-----	-----
G36AN314			28.233
G36AN340			27.587
G36N0137			24.173
G36T0086	158277.213	-7112.893	69.935
G36T0245	158094.537	-1777.912	71.467
G36T0485	162076.134	-1153.516	80.027
G36T0504	164554.883	-1747.450	44.805
G36T0756	165127.497	-4027.942	135.316
G36T0768	169404.041	-1757.135	31.942

GLOBAL TEST

Constrained adjustment: f= 101 Spvv= 9366.35328017

Free adjustment : f= 87 Spvv= 156.60968909

Table value = 1.81 (Kji-square, f=14, alfa=0.0500)

Calculated value = 65.44 ***

There are constraints in the network

LOKALIZATION OF KNOWN CONTROL POINTS WITH ERRONEOUS (INCORRECT) COORDINATES/HEIGHTS

POINT	dSpvv	Distance	Tolerance limit
-----	-----	-----	-----
G36T0086	29.323	0.013	0.072
G36T0245	198.430	0.025	0.072
G36T0485	2077.658	0.057	0.058
G36T0504	9197.163*	0.100	0.011
G36T0756	673.748	0.046	0.091
G36T0768	3381.049	0.098	0.073

Table value =2.70 (Fisher, f1=3, f2=98, alfa = 0.0500)

Incorrect known point could be: G36T0504. The difference between old/new is significant

Comments:

f: degree of freedom $Spvv = \sum pvv$ $dSpvv = \Delta \sum pvv$ (the reduction of $\sum pvv$ compared to the $\sum pvv$ of the original constrained net.

The test failed as the calculated value is bigger than the table value.

The localization of point G36T0504 : $\Delta \sum pvv$ is three times larger than the other $\Delta \sum pvv$, no doubt in this case.

(2) TEST OF COORDINATES AND HEIGHTS

DATUM: NGO1948 - AXIS 3

ADJUSTMENT IN 3-D

KNOWN COORDINATES [meter]

POINT	X	Y	H
G36AN314			28.233
G36AN340			27.587
G36N0137			24.173
G36T0086	158277.213	-7112.893	69.935
G36T0245	158094.537	-1777.912	71.467
G36T0485	162076.134	-1153.516	80.027
G36T0504	164554.983	-1747.450	44.805
G36T0756	165127.497	-4027.942	135.316
G36T0768	169404.041	-1757.135	31.942

GLOBAL TEST

Constrained adjustment: f=	101	Spvv=	170.03707099
Free adjustment :	f= 87	Spvv=	156.60964250

Table value	=	1.81	(Kji-square, f=14, alfa=0.0500)
Calculated value	=	0.53	

CONSTRAINED ADJUSTMENT

GIVEN COORDINATES [meter]

POINT	X	Y	H
G36AN314			28.233
G36AN340			27.587
G36N0137			24.173
G36T0086	158277.213	-7112.893	69.935
G36T0245	158094.537	-1777.912	71.467
G36T0485	162076.134	-1153.516	80.027
G36T0504	164554.983	-1747.450	44.805
G36T0756	165127.497	-4027.942	135.316
G36T0768	169404.041	-1757.135	31.942

Comments:

Before this computation is the coordinates of G36T0504 checked and the x coordinate is corrected with 0.10 m (compare with the x value on the previous page). The alternative was to consider this point as an unknown, but here we found an error, corrected the error and kept the point as a known point.

The global test passed in this computation, no more gross errors on the coordinates and heights

NEW COORDINATES AND STANDARD DEVIATIONS [meter]

POINT	X	Y	H	sX	sY	sH
862	154445.509	-3743.770	3.103	0.002	0.002	0.003
918	152784.003	-3163.958	8.630	0.003	0.002	0.003
2241	159929.341	-713.301	50.031	0.002	0.001	0.003
3167	160183.461	-7161.216	6.945	0.002	0.002	0.005
3171	160261.751	-6512.924	11.040	0.002	0.001	0.004
7001R	157325.310	-1621.067	47.351	0.003	0.002	0.004
7001V	162660.646	2809.892	31.455	0.005	0.003	0.010
7011	165769.272	-424.388	57.543	0.002	0.001	0.004
G36AN314	153458.334	367.477		0.004	0.003	
G36AN340	153742.768	-1803.999		0.003	0.002	
G36N0137	166378.487	-606.350		0.003	0.002	
GPS3A	158306.749	-5803.434	2.913	0.002	0.001	0.003
GPS4	162524.716	-5232.930	0.846	0.005	0.003	0.009
GPS6	165690.120	-1542.890	27.549	0.002	0.001	0.004
GPS10	160952.904	-3440.797	2.087	0.001	0.001	0.003
OE36	159093.705	-4185.765	11.497	0.002	0.001	0.003

ADDITIONAL UNKNOWNNS AND THEIR STANDARD DEVIATIONS

	Value	Std. dev.	Test value
Rotation [gon]:	-0.00026	0.00001	-23.63
Scale [m/km]:	-0.0147	0.0002	-71.35
Deflection of the vertical, X [gon]:	0.00036	0.00003	12.54
Deflection of the vertical, Y [gon]:	0.00032	0.00005	6.44

Table value =1.98 (Student-t, f=100, alfa = 0.0250)

Summary

The number of iterations	:	1
The number of observations, bearings	:	50
The number of observations, distances	:	50
The number of observations, height differences:		50
The number of observations	:	150
The number of unknown coordinates :		32
The number of unknown height coordinates	:	13
The number of additional unknowns	:	4
The number of unknowns	:	49
The degree of freedom	:	101
The number of correlations	:	150
The sum of pvv	:	170.03706699
Calculated std. dev. of the unit weight, M0	:	1.2975
Assumed std. dev. Of the unit weight	:	1.0000

TEST of M0

Table value = 125.46 (Kji-square, f=101, alfa=0.0500)
 Calculated value = 170.04 *

Comments:

See that the additional unknowns are estimated in this adjustment. All of them are tested and they are significantly different from zero

B.3 Internal reliability

The problem is: How can the remaining gross errors after a gross error detection is done, corrupt or deform the adjusted coordinates and heights. This is analyzed by internal and external reliability testing (IP and EP).

In a Norwegian dictionary named "Ordbok for kart og oppmåling" from 1989 the internal reliability is defined as "*a measure of how well the observations in a network control each other, that is how a (gross) error in an observation is reflected in the corresponding adjustment correction of the actual observation.*"

By using methods from the *statistics*, the largest possible (gross) error on an observation, without being detected, is calculated. If the network is good, the possible remaining errors will be small, the network's ability to reveal a gross error is large. Where the network is poorly structured, the remaining errors will become larger. See also chapter B.1.1.

The accuracy of the estimated coordinates and heights in a network cannot be accepted by setting up tolerances of the values of the internal reliability. Internal reliability does not indicate the accuracy or the reliability of the coordinates. In the internal reliability we can compute the largest (possibly) remaining errors on the observations. These remaining errors are used in the external reliability to check the possible deformations on the network. *NOTE: Possible errors, as we are using mathematical statistics to calculate the remaining errors!*

The definition of internal reliability states it is the *controllability* of the observation we want to examine. Therefore, the internal reliability is normally based on a *free network* and *free adjustment*. But in the *external reliability* we use the calculated statistical largest gross errors of a *constrained* network.

B.3.1 Measures of internal reliability

There are several measures of internal reliability:

1. Redundancy number
2. Largest remaining gross errors
3. Maximum relative gross error
4. The minimum detectable gross error (MDE)

(1) Redundancy:

Redundancy, r : The proportion of the gross error as reflected in the adjustment correction, v .

$$r = \frac{-v}{\nabla}$$

The minus sign in the formula is owing to the definition of the formula. This formula applies to independent variables, and the adjustment corrections, v , are not independent....

We make a simplification here, we assume they are independent. Then the values of the redundancy number varies between 0 and 1: $0 < r_{ii} < 1$

Properties of the R matrix:

The redundancy matrix, $\mathbf{R}$, is a square, not symmetric and a singular matrix.

The elements in the main diagonal, r_{ii} , of the redundancy matrix normally has values between 0 and 1. The numerical value of the elements shows how large share of the gross error of observation

no i is shown in the adjustment correction of observation no i , v_i . Observations with low redundancy will display a small portion of its gross error in the adjustment correction of the actual observation, and thus making it difficult to reveal the gross errors in a gross error test.

$\sum r_{ii} = n - e \Rightarrow$ the sum of the elements in the main diagonal is equal to the number of excess measurements (the degree of freedom).

The elements outside the main diagonal, r_{ij} , normally has values between 0 and 0.5. The numerical value of the elements shows in which degree a gross error in observation no i is shown in the adjustment correction of observation no j , v_j . It is therefore not certain that the gross error is at the observation with the largest adjustment correction. This occurs when r_{ii} is less than r_{ij} , and it means we cannot use the adjustment corrections to detect gross errors.

Some formulas for the R-matrix:

$$\begin{aligned} \mathbf{R} &= \mathbf{Q}_{vv} \mathbf{P} \\ \mathbf{R} &= \mathbf{I} - \mathbf{A}(\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P} = \mathbf{I} - \mathbf{A} \mathbf{Q}_{xx} \mathbf{A}^T \mathbf{P} \\ \text{and } \mathbf{R} &= \mathbf{R} \mathbf{R} \text{ (idempotent matrix).} \end{aligned}$$

Observations that cannot be controlled by other observations are defined as *non-testable observations*. Examples of such non-testable observations are:

- Polar measurements, both with total station and GPS.
- Orientation directions in known point to other known point by total station measurements.

If the redundancy number is less than 0.005, the actual observation is regarded as not testable.

(2) Largest remaining gross error

Definition of the remaining gross error: A remaining gross error is an error that is not large enough to be characterized with sufficient certainty as a gross error of an observation.

NB: Remember that the gross error from a statistical testing is a statistical value and it is defined as an error which is too small or too uncertain to be considered as an observation with a gross errors (threshold size).

For each of the observations and their remaining errors a confidence interval is calculated. The boundaries of the confidence interval are defined as follows:

$$\left[\hat{\nabla} - \hat{\sigma}_{\nabla} \cdot T_{\alpha/2, f}, \hat{\nabla} + \hat{\sigma}_{\nabla} \cdot T_{\alpha/2, f} \right]$$

$\hat{\nabla}$: Estimated gross error, see the gross error test

$\hat{\sigma}_{\nabla}$: Estimated standard deviation of gross error, see the gross error testing

$T_{\alpha/2, f}$: Student-t table value with f degree of freedom and test size α

α : Confidence level or test size, common choice is 5%

The *largest remaining gross error*, ∇_0 , is defined as one of the two boundary values with the highest *absolute value*. See example exercise no 9 for an example.

For more about the largest remaining gross error, see chapter B.1.4.

(3) Maximum relative gross error (MRG)

MRG is defined as the largest remaining gross error in multiples of the standard deviation of the observation:

$$\frac{\nabla_{0i}}{\hat{\sigma}_i}$$

If the maximum relative gross error has a value of 3, we can argue that the actual observation does not contain a gross error greater than three times the observation own standard deviation, with the selected significance level.

Generally: We can have a remaining gross errors that may be about 7 to 8 times the standard deviation without being detected by the statistical testing, see for instance example exercise no 9 (gross error).

(4) The minimum detectable gross error (MDF)

MDF is associated with data snooping as a method to determine gross errors. This method of Baarda's is not syllabus for the course Geodesy, and will not be discussed here. See B.1.2.

B.3.2 Example, network A and B. Internal reliability. 2-D test.

Two new control points (P and Q) are measured by using a total station. Only coordinates on the plane, x and y, are of interest, not the heights: It is a 2-dimensional case. The observations are horizontal directions, distances in the plane. The 3rd observation type, orthometrical height differences, is not included in the adjustments.

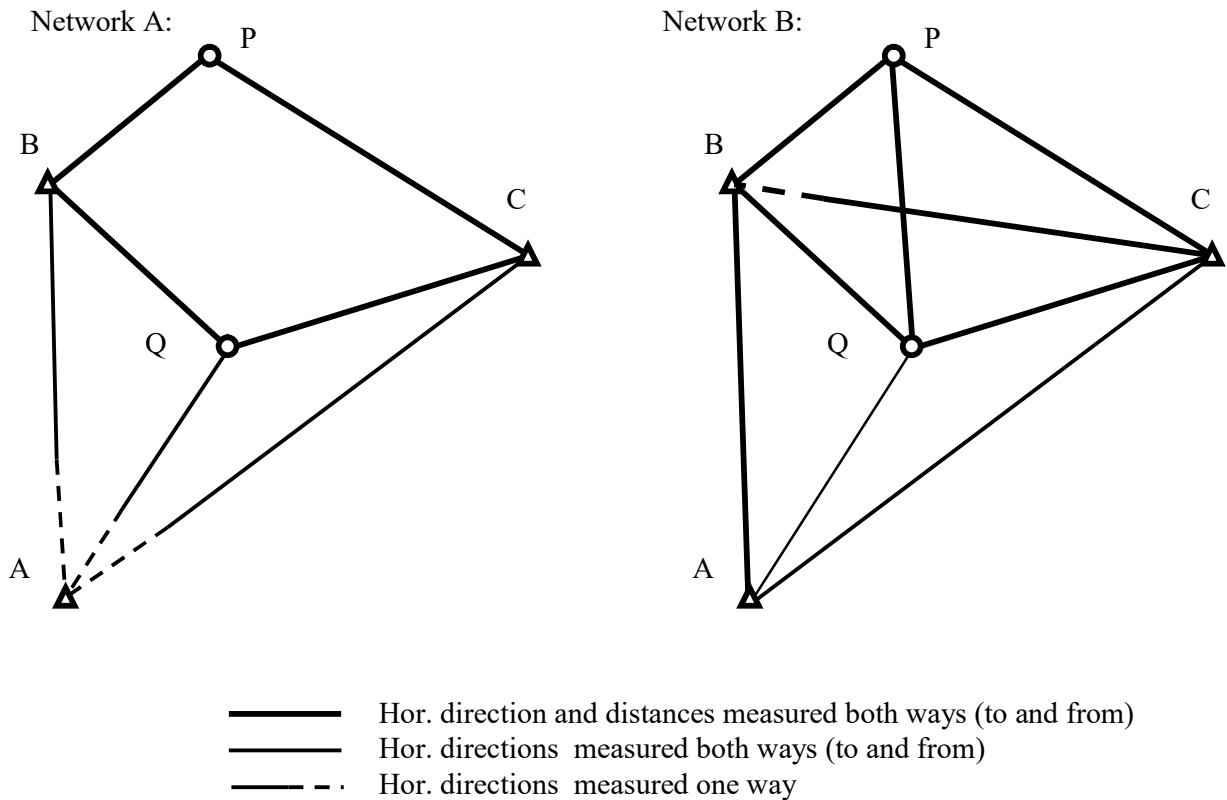


Figure B.9: Network A and B. Observations

Adjustment no A: In the adjustment of network A the scale was fixed (not treated as an unknown).

Adjustment no B: The reliability analysis of network A showed that it was necessary with additional measurements to meet the tolerances concerning the accuracy of the coordinates and heights of P and Q. Therefore some extra observations were added to network A (simulated observations), and an additional adjustment was carried out, adjustment no B.

Note that reliability analysis can be done with real and/or simulated observations. We can therefore carry out a pre-test of the planned network and check if the tolerances are met.

The networks were computed using the land surveying program VG-Land (now named GIS/Line), and the results of the internal and external reliability for the two calculations are shown below.

Internal reliability in network A and B:

We will evaluate the result of the internal reliability (IP) in the two calculations A and B, both separately and compared with each other. Both redundancy and the largest remaining gross errors are used in the analysis.

Key words:

- IP for distance measurements,
- IP for horizontal direction measurements,
- improvements in the network from A to B.

B.3.3 Explanations to the terms used in the IP calculations (VG-Land, 2004):

Column 1: From point (the observation start in this point)

Column 2: Observation set number (all have value 1 here, measured at the same time)

Column 3: To point (the observation ends in this point)

Column 4: Observation type: R: Horizontal direction, D: Distance

Column 5: Redundancy number (elements on the main diagonal, r_{ii})

Column 6: Estimated gross errors

Column 7: Internal reliability, is the largest remaining gross error

Column 8: Warnings about low redundancy numbers, indicated by *

***: $r_{ii} < 0.10$ means that redundancy number is less than 0.10

** : $0.10 < r_{ii} < 0.20$

* : $0.20 < r_{ii} < 0.30$

B.3.4 Internal reliability of network A

FREE NETWORK, FREE ADJUSTMENT

Here the method based on the confidence interval is used, and not based on the student-T testing which is the alternative or option.

INTERNAL RELIABILITY - Confidence Interval [m / gon]

From	To	Redundancy	Est.gross error	Int reliab.

B	1 A	R 0.05	-0.00536	-0.00897***
B	1 P	R 0.31	0.00114	0.00503
B	1 P	D 0.03	-0.062	-0.253 ***
B	1 Q	R 0.31	-0.00019	-0.00455
B	1 Q	D 0.07	-0.033	-0.203 ***
C	1 A	R 0.40	-0.00197	-0.00329
C	1 P	R 0.52	0.00109	0.00384
C	1 P	D 0.13	0.012	0.191 **
C	1 Q	R 0.53	0.00044	0.00370
C	1 Q	D 0.08	0.044	0.218 ***
P	1 B	R 0.33	-0.00043	-0.00461
P	1 C	R 0.33	0.00043	0.00461
Q	1 A	R 0.20	0.00280	0.00468*
Q	1 B	R 0.36	-0.00073	-0.00457
Q	1 C	R 0.36	-0.00082	-0.00459

Table value = 3.18 (Student-t, f=3, alfa = 0.0250) e.g.:
alfa/2, two tailed test

Summary

The number of observations, hor. directions:	11
The number of observations, distances	4
The total number of observations	15
The number of unknown coordinates	10
The number of unknown orientation elements (plane bearings):	4
The number of unknowns	14
Rang defect	3
The corrected number of unknowns	11
The number of excess measurements (degree of freedom):	4

Comments:

Adjustment no A:

Low value of redundancy number for the direction B-A: Can have a very large remaining gross error.

The distance B-P is also critical with respect to gross errors. Total number of redundancy, which is equal to the degree of freedom: $f = n - e = 4$.

B.3.5 Internal reliability of network B

FREE NETWORK, FREE ADJUSTMENT

Here the method based on the confidence interval is used, and not based on the student-T testing which is the alternative or option.

INTERNAL RELIABILITY - Confidence Interval [m / gon]

From	To	Redundancy	Est.gross error	Int reliab.	

A	1 B	R 0.61	-0.00022	-0.00265	
A	1 B	D 0.07	-0.037	-0.241	***
A	1 C	R 0.56	0.00105	0.00348	
A	1 Q	R 0.61	-0.00073	-0.00310	
B	1 A	R 0.40	0.00108	0.00396	
B	1 P	R 0.49	0.00082	0.00346	
B	1 P	D 0.08	-0.119	-0.188	***
B	1 Q	R 0.54	-0.00155	-0.00389	
B	1 Q	D 0.23	0.005	0.081	*
C	1 A	R 0.68	-0.00114	-0.00330	
C	1 B	R 0.73	-0.00111	-0.00318	
C	1 P	R 0.71	0.00103	0.00316	
C	1 P	D 0.19	-0.006	-0.118	*
C	1 Q	R 0.70	0.00122	0.00332	
C	1 Q	D 0.11	0.003	0.127	**
P	1 B	R 0.55	0.00008	0.00263	
P	1 C	R 0.58	-0.00070	-0.00314	
P	1 Q	R 0.63	0.00057	0.00292	
P	1 Q	D 0.29	0.044	0.115	*
Q	1 A	R 0.38	0.00066	0.00370	
Q	1 B	R 0.63	-0.00145	-0.00361	
Q	1 C	R 0.57	-0.00068	-0.00315	
Q	1 P	R 0.67	0.00156	0.00359	

Table value = 2.23 (Student-t, f=10, alfa = 0.0250) e.g.:
 alfa/2, two tailed test

Summary

The number of observations, hor. directions:	17
The number of observations, distances :	6
The total number of observations :	23
The number of unknown coordinates :	10
The number of unknown orientation elements (plane bearings):	5
The number of unknowns :	15
Rang defect :	3
The corrected number of unknowns :	12
The number of excess measurements (degree of freedom):	11

Comments: Improvements for all observations, low redundancy number on the distance A-B, which easily can be explained by referring to a better geometry and a larger number of observations. Still the distance B-P has the largest remaining gross error. $f = n - e = 11$.

But: Still some asterisks in the test, the network is not OK.

The control point P is critical. There should have been measured to P from a fixed control point north of P. Then point P will be within a polygon drawn between the known control points. As seen in the deformations above, it is especially the distance B-P which is critical for the results of external reliability analysis. See the results of the external reliability for more, chapter B.6.1.

B.4 External reliability, ER

External reliability is a measure of the impact on the coordinates and heights of the potential remaining (gross) error in the observations.

It is an investigation of the potential deformation of the network of control points. External reliability is a deformation analysis: The network's ability to avoid deformations of the possible remaining gross errors is tested.

The basis for calculating the external reliability is the largest remaining gross error for each observation. See the internal reliability for this type of error.

It is the geometry of the network, the accuracy (weight) of the observations and the correlations between the observations that determine to what extent a possible remaining gross errors deform / affects the unknown coordinates and heights. A good network and a low weight on an actual observation give the effect that the gross error of this observation is small.

A test on external reliability is performed on a *constrained network*, i.e. by holding the control points which have known coordinates and heights fixed. By using a constrained network instead of a free network any remaining errors (or the accuracy of the known control points) also will affect the outcome of the external reliability. This effect is in many cases a desired effect.

In the external reliability test we also calculate the additional unknowns of the network, as it is done in the final adjustment of the network. The additional unknowns are the unknown scale, the rotation and the components of deviation of vertical (north and east).

Constrained network and internal reliability:

Since we use a constrained network, the calculation of the largest remaining gross errors in the internal reliability test also should be made in a constrained network (and not in a free network, which is the ideal situation). By using constrained network both for internal and external reliability, we are using the numerical values of the gross errors that are related to the situation for the constrained network.

We obtain different values for gross errors and redundancy in a constrained network compared to the values in a free network. As an example, if the distance between two known point is measured: In a constrained network where the two known points are kept fixed, the redundancy number of this distance will be equal to 1. The proportion of the gross error as reflected in the adjustment correction, v is in this case equal to 1.

Formula from the definition of redundancy:

$$r = \frac{-v}{\nabla}$$

B.4.1 The method

It is important to emphasize that this is a *statistical* test, where we use a *possible* remaining error in the observations.

1. Are there n observations in the network, the network is calculated n times, with only *one* observation with its possible gross error added, in each calculation.

2. The specific adjustment containing the maximum deformation (the maximum change in values) of, for example, the height of a point, is used for the deformation analysis.
3. The obtained deformations in point 2 is compared with the limits of the Norwegian "Geodata standard". See examples below.

B.4.2 The Norwegian "Geodata standard"

"Kvalitetssikring av oppmåling, kartlegging og geodata. ("Geodatastandarden") Quality assurance of land surveying, mapping and geodata (the Geodata standard)			
"Grunnlags-nett" Fundamental network	"Kart og kartdata" Maps and map data	"Eiendoms- informasjon" Real estate information	"Plassering og beliggenhets-kontroll" Control of placing and location

Figure B.2. Examples of standards referring to the Geodata standard.

The Geodata standard sets tolerances of to the maximum values for the various types of deformations in the external reliability:

- *Level 1: Point deformation.* Highest allowed deformations (changes) on the coordinates and heights and on the total deformation of the point.
- *Level 2: Deformation of the plane bearing, the distance and the height differences between neighbouring points:* Highest allowed deformations (changes) on the plane bearing, the distance and the height differences to the nearest points.
- *Level 3: Deformation of the horizontal angles and the scale factors:* Highest allowed deformations (changes) on the horizontal angle between and on the scale to two of the nearest points of the control point we are checking.

Point deformation is normally used in measurements of real property /estates. For fundamental networks and control points the tolerances of the other two deformations are in use. See the numerical examples for more on this topic.

The values in the last deformation above (level 3), gives the best analytical results for the YP. It can reveal some local inhomogeneous parts of the network than the other two levels cannot do.

NB, important: With neighbouring points we mean neighbouring points in general, whether it is measured between them or not. This is an important feature of deformation analysis: Although it is not measured between the actual neighbouring points, it is probable it will be measured between them in the future. Therefore it is important to include all the neighbouring points in the actual adjustment project, whether it is measured between them or not.

The computed deformations are compared with tolerance values from the Geodata standard (the part Fundamental networks). There you will find details and numerical values of the tolerances. Pages with some key topics are included below.

We can choose the quality class we want to compare the deformations of the network with. There are four classes (1 to 4), see tables B.3a (in Norwegian) and B.3b (in English). In addition, the Norwegian Mapping Authority's has its own tolerance requirements for their fundamental networks.

The threshold value Δ in the formula of the tolerances is a calculated value based on the current

geometry between the three points involved (the point of interest and two of the neighbouring points). This threshold value Δ will therefore vary from point to point, and between pairs of directions to neighbouring points, since the distances between the 3 points are included in the formula.

The requirements are an upper limit for error, the tolerance.

Since this is a statistical test, a change in the numerical values of the observations will not give lower values of the deformation. We must make additional measurements to obtain a better geometry, if the requirements of the tolerances are not met.

B.4.3 Some pages from the Geodata standard

The following pages are copied from the standard, and translated:

(1) In the plane, tolerances

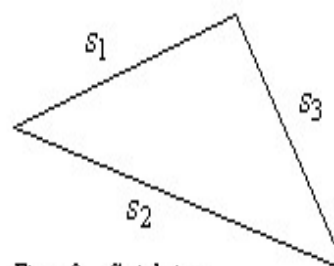
Formulas:

k – a constant related to the individual point

p – a constant related to all angle differences and scale differences

For two lines s_1 and s_2 between a point and any two other points with a distance s_3 between them, the tolerances of the *horizontal angle differences* and the *scale factors* are calculated by the following formula:

$$\Delta = \sqrt{p^2 + \frac{s_1^2 + s_2^2 + s_3^2}{s_1^2 \cdot s_2^2} k^2}$$



Figur 2. Se teksten.

Figure B.10. Formula and figure

In the formula the distances s_1 , s_2 , s_3 are in km, p in ppm (parts per million, for example mm per km for a distance), k in mm.

The numerical values of p and k are chosen from the actual quality class, see tables B.3a (in Norwegian) and B.3b (in English).

	Parameter, grunnriss _	p	k
Klasse	Punkter i	(ppm)	(mm)
	Geodetisk stamnett	3	3
	Geodetisk landsnett	6	6
1	Områdetype 1 Byområde	10	10
2	Områdetype 2 Tettbygd/utbyggingsområder	20	20
3	Områdetype 3 Spredtbygd/dyrket mark/skog	50	50
4	Områdetype 4 Fjell/ekstensiv arealutnytting	100	100
5	Ukjent kvalitet, eller dårligere enn klasse 4		

Table B.3a. In Norwegian. Values of the constants p and k for the calculation in the plane

	Parameter, in the plane	p	k
Class	Points in	(ppm)	(mm)
	Geodetic network, “ stamnett ”. National fundamental points	3	3
	Geodetic network, “ landsnett ”. First order net of the municipalities	6	6
1	Areas of type 1 Urban area	10	10
2	Areas of type 2 Densely populated/development area	20	20
3	Areas of type 3 Rural area/cultured field/forest	50	50
4	Areas of type 4 Mountainous area/extensive development	100	100
5	Unknown quality, or poorer than class 4		

Table B.3b. In English. Values of the constants p and k for the calculation in the plane.

Example:

For $s_1 = s_2 = s_3 = 1$ km, what is the tolerances in the 4 area types 1-4?

(ppm: parts per million, for instance mm per km for a distance). Answer:

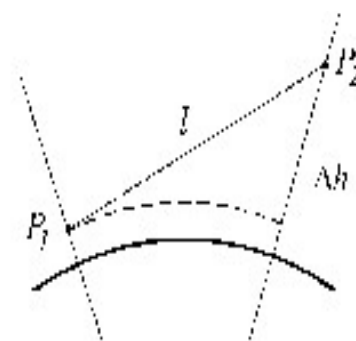
Quality class	1	2	3	4
Tolerance (ppm)	20	40	100	200
In mm:	20 mm	40 mm	100 mm	200 mm

Table B.4

(2) The heights, tolerances

The height differences between two points. The *tolerances for the height differences* are given in ppm (parts per million). The tolerance is calculated by the following formula:

$$\Delta = \sqrt{p^2 + 2 \frac{k^2}{l^2}}$$



Figur 3. Se teksten.

Figure B.11. Formula and figure

In the formula above are p in ppm (parts per million) and k in mm. The slope distance, l , is in km (NB: the shortest slope distance, not along the line of the levelling if levelling is used).

The numerical values of p and k are chosen from the actual quality class.

These values are given in the tables below. See tables B.5a (in Norwegian) and B.5b (in English) .

	Parameter, høyde _	p	k
Klasse	Punkter i	(ppm)	(mm)
	Geodetisk stamnett , ellipsoidiske høyder	3	3
	Geodetisk stamnett , høyder i nasjonalt system	Følger områdetype	
	Geodetisk landsnett , ellipsoidiske høyder	6	6
	Geodetisk landsnett , høyder i nasjonalt system	Følger områdetype	
	Presisjonsnivellment , høyder i nasjonalt system	5	5
1	Omr.type 1 Byområde, høyder i nasj. syst.	5	5
2	Omr.type 2 Tettbygd/utbyggingsområder, høyder i nasj. syst.	20	20
3	Omr.type 3 Spredtbygd/dyrket mark/skog, høyder i nasj. syst.	50	25
4	Omr.type 4 Fjell/ekstensiv arealutnytting, høyder i nasj. syst.	100	30
5	Ukjent kvalitet, eller dårligere enn klasse 4		

Table B.5a. In Norwegian. Values of the constants p and k for the height tolerances

	Parameters, heights	p	k
Class	Points in	(ppm)	(mm)
	Geodetic " stamnett ", ellipsoidal heights	3	3
	Geodetic " stamnett ", heights in the national height system	Use the area type	
	Geodetic " landsnett ", ellipsoidal heights	6	6
	Geodetic " landsnett ", heights in the national height system	Use the area type	
	Precision levelling , heights in the national height system	5	5
1	Area type 1 Urban area, heights in nat. syst.	5	5
2	Area type 2 Densely populated/development area, heights in nat. syst.	20	20
3	Area type 3 Rural area/cultured field/forest, heights in nat. syst.	50	25
4	Area type 4 Mountainous area/extensive development, heights in nat. syst.	100	30
5	Unknown quality, or poorer than class 4		

Table B.5b. In English. Values of the constants p and k for the calculations of the height tolerances

Example:

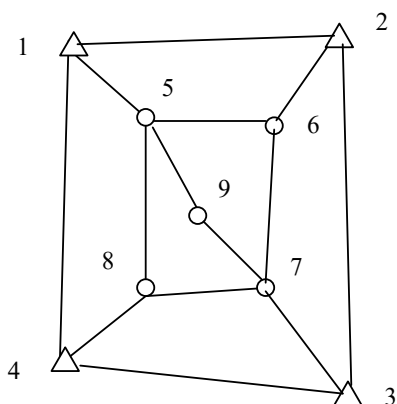
For the slope distance, l , the tolerances in the different area types are calculated by using the formula above:

Area type:	1	2	3	4
Tolerance (ppm) $l=100$ m:	71	284	357	436
Tolerance (ppm) $l=1$ km:	9	35	61	109

Converted to height differences:

Area type:	1	2	3	4
Tolerance (in mm) $l=100$ m	7	28	36	44
Tolerance (in mm) $l=1$ km	9	35	61	109

B.5 Numerical example, GPS network. External reliability



The GPS network is used as an example in chapter 9.3 in the textbook in basic land surveying "Grunnleggende landmåling". For a copy of the figure showing the observations, see to the right. The network is analyzed in an analysis program. Suchlike data programs are part of map and adjustment programs in commercial land surveying applications. We choose control point 9 as the point of interest, there are similar calculations for the other new control points. The tolerances are from quality class 2.

Note that all the neighbouring points are tested, regardless of it is measured between them or not. This is the reason for the control points 1, 6 and 8 are included in the tables below, although it is not measured between point 9 and these points.

Figure B.12 GPS network

- **Height deformation.** Of all the possible height differences from control point 9, it is the height difference to the known control point 1 which is most deformed. The reason for this deformation is a possible gross error on the measured zenith angle from point 9 to point 7, see figure B.12. A height deformation of 20.70 ppm in table B.6 is equal to 0.064 m for the distance 9-1 (3114.5 m)

Point, from	Point, to	Obs.	Gross error on the obs.	Deformation of the ΔH	The Geodata standard. Tolerance
9	1	z_{9-7}	0.00190 gon	20.70 ppm = 0.064 m	22.75 ppm = 0.071 m

Table B.6 Height deformation

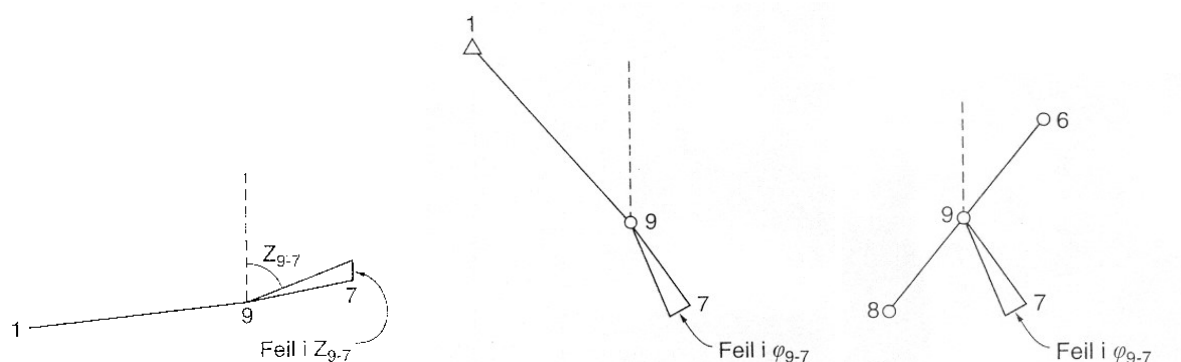


Figure B.13 Height deformation. Figure B.14 Angle deformation. Figure B.15 Scale deformation
The text "Feil i ..." means Error in observation

- **Angle deformation.** Of all the possible angles in control point 9 it is the angle between the control points 1 and 7 that is most deformed. The reason for this deformation is a possible gross error in the measured plane bearing from point 9 to point 7, see figure B.14. An angle deformation of -19.22 ppm in table B.7 is equal to -0.00122 gon, as 1 ppm (parts per million) is equal to 10^{-6} radians.

Point, from	Point 1, to	Point 2, to	Obs.	Gross error on the obs.	Deformation	The Geodata standard. Tolerance
9	1	7	ϕ_{9-7}	-0.00128 gon	-19.22 ppm	33.87 ppm

Table B.7 Angle deformation

- **Scale deformation.** By comparing the scale (the scale factors) to all neighbouring points in control point 9, the greatest difference in scale is between the distances 9-6 and 9-8. The reason for this deformation is the possible gross error in the measured plane bearing from control point 9 to point 7. The gross error displaces point 9 in such way that the distances from 9 to 6 and 8 are deformed in scale. A scale deformation of 19.49 ppm is equal to 19.49 mm per km length. See figure B.15 and table B.8.

Point, from	Point 1, to	Point 2, to	Obs.	Gross error on the obs.	Deformation	The Geodata standard. Tolerance
9	6	8	ϕ_{9-7}	-0.00128 gon	19.49 ppm	37.74 ppm

Table B.8 Scale deformation

B.6 Example. Network A and B, total station. External reliability

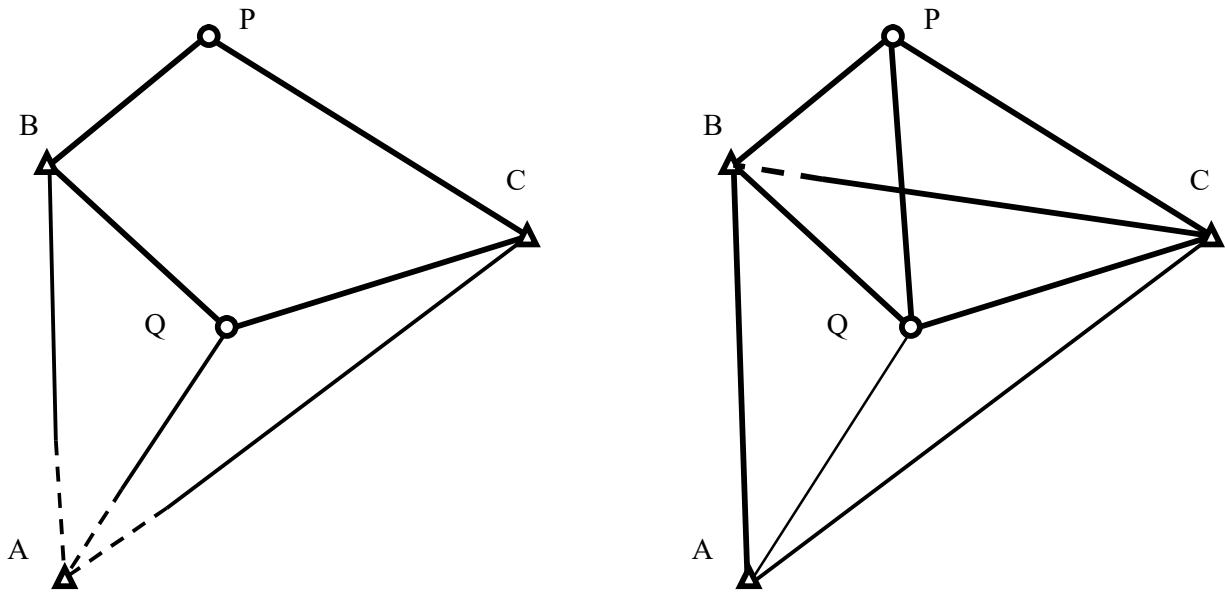


Figure B.16 Network A and B. Totalstation.

We use the same networks as in the example of internal reliability in chapter B.3.2, see the calculations A and B of the same networks.

The networks were calculated by using the data program VG-Land (now GIS/Line). The results of the external reliability test of the two calculations A and B are shown in chapter B.6.2 and 3 in English and B.6.4 and 5 in Norwegian.

External reliability, the following are examined in this example:

1. What is the *maximum point deformation* on the two estimated control points P and Q?
Comment on the improvements from the calculation A to calculation B.

2. What is the maximum deformation of the horizontal direction and of the distance in the 5 control points of the network?
 Comment on the improvements from the calculation A to calculation B.
 Interpret the result in the control point P, also with a figure. Which observation is critical for the deformation?
- 3: What is the *maximum deformation of the horizontal angle and the scale* in the 5 control points of the network?
 Comment on the improvements from the calculation A to calculation B.
 Interpret the results in control point P, preferably with a figure! Why is the observation 2 (see the bottom of the printout from the computations) the critical observation concerning the deformations?
- 4: Calculate the tolerance (threshold value) in the point of interest C, where values of p and k from quality class 2 (second best class) are to be used.
 Evaluate the tolerances compared with calculated values calculated in the printout.
- 5: Make a brief account (conclusion) based on the results of tests of the external reliability of the two calculations A and B.

B.6.1 Comments on the external reliability tests of the network A and B

See especially after * markings in the right hand column. An asterisk, *, indicates that the tolerance requirements of the Geodata standard is exceeded, where *** is very large excess.

1. Maximum point deformation:

Point P: As expected (?) it is the distance observation B-P which gives the biggest deformation in point P in both the calculations A and B. The numerical values are reduced in B compared to A, but still large deformation.

Point Q: The deformation is reduced from A to B, both for the coordinates x and y , and total (the value of P in the calculations).

2. Maximum deformation of the horizontal direction and of the distance:

See the printouts for values and difference in values from A to B.

For point of interest P:

Max deformation in horizontal direction from P to C is caused by the distance B-P. Max deformation on the distance from P to B because of the distance B-P. The same points are included in both the calculations A and B.

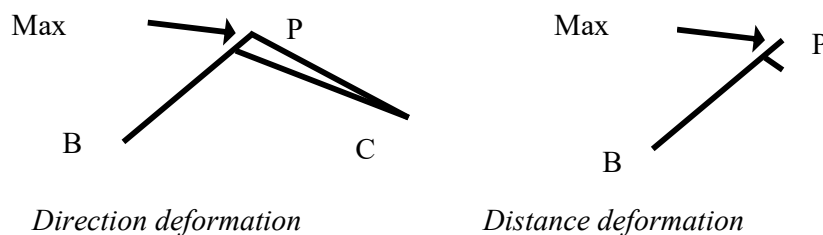


Figure B.17 Deformations

3. Maximum deformation of the horizontal angle and the scale:

See the printouts for values and difference in values from A to B.

In point P: The networks A and B: *The difference in scale* is biggest for the distances to the points B and C. The reason is again the observed distance B-P. The difference in scale between the two distances is given in *ppm* which is mm per km.

Network A, in point P: The *angle deformation* is biggest for the angle between the points B and C.

Network B, in point P: The *angle deformation* is biggest for the angle between the points B and Q.

In both the networks, the distance B-P again is the critical observation concerning the deformations in point P.

Point Q: See also network B, it is the point Q which gives the largest deformation of the angle.

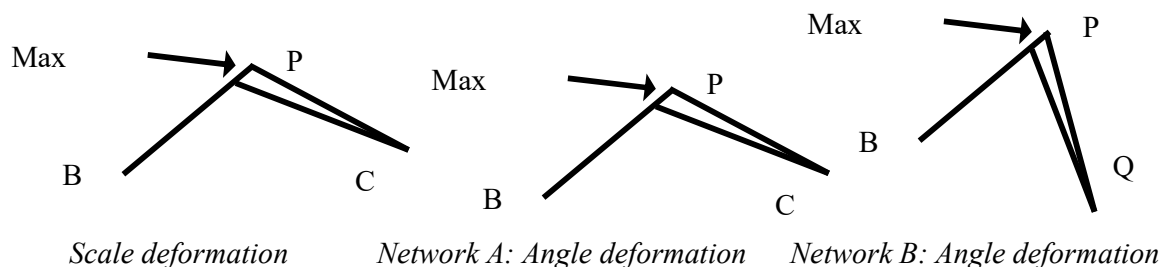


Figure B.18a, b and c Deformations

4. The Geodata standard

Tolerances (threshold values) in the point of interest C (class 2):

First the value of Δ is calculated, see the formula in chapter B.10. Notice that the same points are involved in both the networks A and B.

NB: There are different values of the tolerances, depending on the 3 points that form the triangles.

Tolerances in the point of interest C (class 2), compare with the values in the printout:

Triangle	Δ value (class 2)	Δ value converted to gon (grads)
CPB	20,64 ppm	0,00132 gon
CQP	21,25 ppm	

5. Conclusion

The additional measurements in B have improved the network, but there are still observations marked * in the external reliability, indicating that the network still does not meet the tolerances of class 2 in the Geodata standard ...

To satisfy the requirements of the Geodata standard, class 2, the control point P is critical. There should have been measured to P from a fixed control point north of P. Then point P will be within a polygon drawn between the known control points. As seen in the deformations above, it is especially the distance B-P which is critical for the results of external reliability analysis.

B.6.2 Explanations of the values in an external reliability test:

The examples below are from network A, see also similar explanations in the internal reliability test.

(1) EXTERNAL RELIABILITY - COORDINATES [meter]

COORDINATE	Observation..[meter/gon]	Int. rel.	Extern. rel.
X P	B 1 P	D	-0.150 -0.106**
Y P	B 1 P	D	-0.150 -0.094*
P P	B 1 P	D	-0.150 0.142**

Column 1: Looking at either the x or y coordinates, or total (P)

The next columns show that it is the distance observation B-P which is the critical observation (distance: D in column 6).

The internal reliability is the maximum remaining gross errors. NOTE: Here calculated in constrained networks, obtaining different values than in the example internal reliability test (where a free network was used).

External reliability is the maximum deformation in meters.

(2) EXTERNAL RELIABILITY - DIRECTION, DISTANCE, HEIGHT DIFFERENCES

[gon / m / km]

Connection	Observation	..[m / gon]	Internal rel.	Extern. rel.
D B	P B	P D	-0.150	-0.050 **
R C	P B	P D	-0.150	-0.00168*

Column 1 indicates whether it is the direction (R), distance (D) or difference in height (dH) that is being examined.

Column 2 and 3 indicate the start and end point (the line, the connection)

Column 4, 5 and 6 indicates the critical observation between two points (here the distance B-P), which gives the max deformation.

Internal reliability is the maximum remaining gross errors

External reliability is the maximum deformation

(3a) EXTERNAL RELIABILITY – SCALE DIFFERENCES [ppm]

Point	Connection 1	Connection 2	Tolerance	Ext. rel.
Observation....[meter/gon]	Internal rel.			
P B	C	23.38	55.26 **	
D B	1 P	-0.150		

(3b) EXTERNAL RELIABILITY – ANGLE DEFORMATIONS [ppm]

Point	Connection 1	Connection 2	Tolerance	Ext. rel.
Observation....[meter/gon]	Internal rel.			
P B	C	23.38	-32.20 *	
D B	1 P	-0.150		

Column 1: "Point" is the point of interest, the point we investigate the deformation in.

Column 2: Connection 1 and 2 are the two other points (relations) that together with the point of interest form the triangle with the biggest deformation from the point of interest.

Line 2: The critical observation, here the distance between B and P. It is the gross error in this observation that gives the maximum deformation.

Internal reliability is the maximum remaining gross errors.

Tolerance is the requirement of the standard, calculated in the triangle between the three points P, B and C in the above example.

External reliability is the calculated deformations of the scale (M) and horizontal angle (V) in the point of interest for the network.

An asterisk, *, in the last column indicates that the deformation is larger than the tolerances in the Geodata standard for the current class.

B.6.3 Example, external reliability. Network A. In English:

NB: The gross errors are calculated by using the confidence interval method.

ADJUSTMENT IN THE PLANE

F

KNOWN COORDINATES [meter]

POINT	X	Y	H
A	631570.304	-2213.195	100.000
B	637363.297	-2595.198	200.000
C	636513.510	4014.202	150.000

NEW COORDINATES AND ESTIMATED STANDARD DEVIATIONS [meter]

Point	X	Y	H	sX	sY	sH
P	639236.582	-513.590		0.013	0.012	
Q	635014.547	87.836		0.015	0.011	

EXTERNAL RELIABILITY - COORDINATES [meter]

COORDINATE	Observation...[meter/gon]			Internal rel.	Extern. rel.

X P	B	1 P	D	-0.150	-0.106**
Y P	B	1 P	D	-0.150	-0.094*
P P	B	1 P	D	-0.150	0.142**
X Q	B	1 Q	D	0.074	-0.054*
Y Q	C	1 Q	D	0.099	-0.060*
P Q	C	1 Q	D	0.099	0.080*

Table value = 2.45 (Student-t, f=6, alfa = 0.0250)

EXTERNAL RELIABILITY - DIRECTION, DISTANCE AND HEIGHT DIFFERENCES [gon / m/km]

Connection	Observation...	[m / gon]	Internal rel.	External rel.
R A Q B Q D			0.074	0.00085
D A Q C Q D			0.099	-0.019
R A P B P D			-0.150	-0.00056
D A P B P D			-0.150	-0.016
R B Q C Q D			0.099	0.00142
D B Q B Q D			0.074	0.016
R B P C P D			-0.090	0.00139
D B P B P D			-0.150	-0.050 **
R Q A B Q D			0.074	0.00085
D Q A C Q D			0.099	-0.019
R Q B C Q D			0.099	0.00142
D Q B B Q D			0.074	0.016

R	Q	C	B	Q	D	0.074	-0.00093
D	Q	C	C	Q	D	0.099	0.018
R	Q	P	B	P	D	-0.150	-0.00160
D	Q	P	B	P	D	-0.150	-0.021
R	C	Q	B	Q	D	0.074	-0.00093
D	C	Q	C	Q	D	0.099	0.018
R	C	P	B	P	D	-0.150	-0.00168*
D	C	P	C	P	D	-0.090	-0.011
R	P	A	B	P	D	-0.150	-0.00056
D	P	A	B	P	D	-0.150	-0.016
R	P	B	C	P	D	-0.090	0.00139
D	P	B	B	P	D	-0.150	-0.050 **
R	P	Q	B	P	D	-0.150	-0.00160
D	P	Q	B	P	D	-0.150	-0.021
R	P	C	B	P	D	-0.150	-0.00168*
D	P	C	C	P	D	-0.090	-0.011

Table value = 2.45 (Student-t, f=6, alfa = 0.0250)

EXTERNAL RELIABILITY - SCALE DIFFERENCES [ppm]

Point	Connection 1	Connection 2	Tolerance	Ext. rel.
Observation....[meter/gon]			Internal rel.	

A	Q	C	20.88	18.78
	D C	1 Q	0.099	
B	C	P	22.18	-50.38 **
	D B	1 P	-0.150	
Q	A	C	23.01	36.69 *
	D C	1 Q	0.099	
C	Q	P	21.25	-18.90
	D C	1 Q	0.099	
P	B	C	23.38	55.26 **
	D B	1 P	-0.150	

Table value = 2.45 (Student-t, f=6, alfa = 0.0250)

Tolerances:

Standard : Geodata standard - network

Quality class: II: Densely populated (p = 20.0 ppm k = 20.0 mm)

EXTERNAL RELIABILITY - ANGLE DEFORMATIONS [ppm]

Point	Connection 1	Connection 2	Tolerance	Ext. rel.
Observation....[meter/gon]			Internal rel.	

A	Q	C	20.88	-13.43
	D B	1 Q	0.074	
B	Q	C	21.28	-22.37 *
	D C	1 Q	0.099	
Q	B	C	23.09	-29.10 *

	D	C	1	Q	0.099	
C		B		P	20.64	-26.43 *
	D	B	1	P	-0.150	
P		B		C	23.38	-32.20 *
	D	B	1	P	-0.150	

Table value = 2.45 (Student-t, f=6, alfa = 0.0250)

Tolerances:

Standard : Geodata standard - network

Quality class: II: Densely populated (p = 20.0 ppm k = 20.0 mm)

STATISTICS

The number of observations, hor. directions : 11
 The number of observations, distance : 4
The number of observations : 15
 The number of unknown coordinates in the plane: 4
 The number of unknown orientation elements (bearings): 4
 The number of unknowns : 8
The number of degree of freedom : 7

The maximum number of points per sector : 3
 The threshold distance for neighbouring points: 10000.000
 The number of single connections : 20
 The number of double connections : 30

B.6.4 Example, external reliability. Network B. In English:

NB: The gross errors are calculated by using the confidence interval method

EXTERNAL RELIABILITY - COORDINATES [meter]

NEW COORDINATES AND ESTIMATED STANDARD DEVIATIONS [meter]

PUNKT	X	Y	H	sX	sY	sH
P	639236.596	-513.590		0.013	0.014	
Q	635014.533	87.828		0.015	0.013	

EXTERNAL RELIABILITY - COORDINATES [meter]

COORDINATE Observation...[meter/gon]				Internal rel.	Extern. rel.
X P	B	1 P	D	-0.134	-0.072*
Y P	B	1 P	D	-0.134	-0.088*
P P	B	1 P	D	-0.134	0.114**
X Q	P	1 Q	D	0.105	-0.034
Y Q	C	1 Q	D	-0.082	0.045
P Q	C	1 Q	D	-0.082	0.055*

Table value = 2.16 (Student-t, f=13, alfa = 0.0250)

EXTERNAL RELIABILITY - DIRECTION, DISTANCE AND HEIGHT DIFFERENCES
[gon / m/km]

Connection		Observation...[m / gon]				Internal rel.	External rel.
R	A	Q	B	Q	D	-0.060	-0.00054
D	A	Q	C	Q	D	-0.082	0.013
R	A	P	B	P	D	-0.134	-0.00057
D	A	P	B	P	D	-0.134	-0.011
R	B	Q	C	Q	D	-0.082	-0.00096
D	B	Q	B	Q	D	-0.060	-0.010
R	B	P	C	P	D	-0.089	0.00104
D	B	P	B	P	D	-0.134	-0.041 *
R	Q	A	B	Q	D	-0.060	-0.00054
D	Q	A	C	Q	D	-0.082	0.013
R	Q	B	C	Q	D	-0.082	-0.00096
D	Q	B	B	Q	D	-0.060	-0.010
R	Q	C	B	Q	D	-0.060	0.00052
D	Q	C	C	Q	D	-0.082	-0.013
R	Q	P	B	P	D	-0.134	-0.00127
D	Q	P	P	Q	D	0.105	0.016
R	C	Q	B	Q	D	-0.060	0.00052
D	C	Q	C	Q	D	-0.082	-0.013
R	C	P	B	P	D	-0.134	-0.00129
D	C	P	C	P	D	-0.089	-0.009
R	P	A	B	P	D	-0.134	-0.00057
D	P	A	B	P	D	-0.134	-0.011
R	P	B	C	P	D	-0.089	0.00104
D	P	B	B	P	D	-0.134	-0.041 *
R	P	Q	B	P	D	-0.134	-0.00127
D	P	Q	P	Q	D	0.105	0.016
R	P	C	B	P	D	-0.134	-0.00129
D	P	C	C	P	D	-0.089	-0.009

Table value = 2.16 (Student-t, f=13, alfa = 0.0250)

EXTERNAL RELIABILITY - SCALE DIFFERENCES [ppm]

Point	Connection 1		Connection 2		Tolerance	Ext. rel.
	Observation....[meter/gon]				Internal rel.	

A		Q		C	20.88	-12.54
	D	C	1	Q	-0.082	
B		Q		P	23.57	-44.63 *
	D	B	1	P	-0.134	
Q		A		C	23.01	-25.31 *
	D	C	1	Q	-0.082	
C		Q		P	21.25	14.73
	D	C	1	Q	-0.082	

P	B	C	23.38	47.85 **
	D B	1 P	-0.134	

Table value = 2.16 (Student-t, f=13, alfa = 0.0250)

Tolerances:

Standard : Geodata standard - network

Quality class: II: Densely populated (p = 20.0 ppm k = 20.0 mm)

EXTERNAL RELIABILITY - ANGLE DEFORMATIONS [ppm]

Point	Connection 1	Connection 2	Tolerance	Ext. rel.
	Observation....[meter/gon]	Internal rel.		
A	Q	P	20.90	-11.60
	D B	1 P	-0.134	
B	Q	P	23.57	-18.92
	D P	1 Q	0.105	
Q	B	P	21.61	-28.01 *
	D B	1 P	-0.134	
C	B	P	20.64	-20.23
	D B	1 P	-0.134	
P	B	Q	22.55	-17.95
	D B	1 P	-0.134	

Table value = 2.16 (Student-t, f=13, alfa = 0.0250)

Tolerances:

Standard : Geodata standard - network

Quality class: II: Densely populated (p = 20.0 ppm k = 20.0 mm)

STATISTICS

The number of observations, hor. directions :	17
The number of observations, distance :	6
The number of observations :	23
The number of unknown coordinates in the plane:	4
The number of unknown orientation elements (bearings):	5
The number of unknowns :	9
The number of degree of freedom :	14
 The maximum number of points per sector :	 3
The threshold distance for neighbouring points:	10000.000
The number of single connections :	20
The number of double connections :	30

B.6.5 Example, external reliability. Network A. In Norwegian:

Ytre pålitelighet, nett A

NB: Grovfeil er bestemt med konfidensintervall-metoden

UTJEVNING I GRUNNRIS

TVUNGEN UTJEVNING

GITTE KOORDINATER [meter]

PUNKT	X	Y	H
A	631570.304	-2213.195	100.000
B	637363.297	-2595.198	200.000
C	636513.510	4014.202	150.000

NYBESTEMTE KOORDINATER MED MIDLERE FEIL [meter]

PUNKT	X	Y	H	sX	sY	sH
P	639236.582	-513.590		0.013	0.012	
Q	635014.547	87.836		0.015	0.011	

YTRE PÅLITELIGHET - KOORDINATER [meter]

KOORDINAT	Observasjon....[meter/gon]			Indre pål.	Ytre pål.
X P	B	1 P	D	-0.150	-0.106**
Y P	B	1 P	D	-0.150	-0.094*
P P	B	1 P	D	-0.150	0.142**
X Q	B	1 Q	D	0.074	-0.054*
Y Q	C	1 Q	D	0.099	-0.060*
P Q	C	1 Q	D	0.099	0.080*

Tabellverdi=2.45 (Student-t, f=6, alfa=0.0250)

YTRE PÅLITELIGHET - RETNING, AVSTAND OG HØYDEFORSKJELL [gon / m/km]

Forbindelse	Observasjon....[meter/gon]					Indre pål.	Ytre pål.
R A Q	B	Q	D			0.074	0.00085
D A Q	C	Q	D			0.099	-0.019
R A P	B	P	D			-0.150	-0.00056
D A P	B	P	D			-0.150	-0.016
R B Q	C	Q	D			0.099	0.00142
D B Q	B	Q	D			0.074	0.016
R B P	C	P	D			-0.090	0.00139
D B P	B	P	D			-0.150	-0.050 **
R Q A	B	Q	D			0.074	0.00085
D Q A	C	Q	D			0.099	-0.019
R Q B	C	Q	D			0.099	0.00142
D Q B	B	Q	D			0.074	0.016
R Q C	B	Q	D			0.074	-0.00093
D Q C	C	Q	D			0.099	0.018
R Q P	B	P	D			-0.150	-0.00160
D Q P	B	P	D			-0.150	-0.021
R C Q	B	Q	D			0.074	-0.00093
D C Q	C	Q	D			0.099	0.018

R	C	P	B	P	D	-0.150	-0.00168*
D	C	P	C	P	D	-0.090	-0.011
R	P	A	B	P	D	-0.150	-0.00056
D	P	A	B	P	D	-0.150	-0.016
R	P	B	C	P	D	-0.090	0.00139
D	P	B	B	P	D	-0.150	-0.050 **
R	P	Q	B	P	D	-0.150	-0.00160
D	P	Q	B	P	D	-0.150	-0.021
R	P	C	B	P	D	-0.150	-0.00168*
D	P	C	C	P	D	-0.090	-0.011

Tabellverdi=2.45 (Student-t, f=6, alfa=0.0250)

YTRE PÅLITELIGHET - MÅLESTOKKDIFFERANSER [ppm]

Punkt	Forb1	Forb2	Normkrav	YP-M.diff.
	Observasjon....[meter/gon]		Indre pål.	
A	Q	C	20.88	18.78
	D C	1 Q	0.099	
B	C	P	22.18	-50.38 **
	D B	1 P	-0.150	
Q	A	C	23.01	36.69 *
	D C	1 Q	0.099	
C	Q	P	21.25	-18.90
	D C	1 Q	0.099	
P	B	C	23.38	55.26 **
	D B	1 P	-0.150	

Tabellverdi=2.45 (Student-t, f=6, alfa=0.0250)

Normparametre:

Normtype : Geodatanormen - Grunnlagsnett
 Normklasse : II : Tettbygd (p= 20.0 ppm k= 20.0 mm)

YTRE PÅLITELIGHET - VINKELDEFORMASJONER [ppm]

Punkt	Forb1	Forb2	Normkrav	YP-V.def.
	Observasjon....[meter/gon]		Indre pål.	
A	Q	C	20.88	-13.43
	D B	1 Q	0.074	
B	Q	C	21.28	-22.37 *
	D C	1 Q	0.099	
Q	B	C	23.09	-29.10 *
	D C	1 Q	0.099	
C	B	P	20.64	-26.43 *
	D B	1 P	-0.150	
P	B	C	23.38	-32.20 *
	D B	1 P	-0.150	

Tabellverdi=2.45 (Student-t, f=6, alfa=0.0250)

Normparametre:

Normtype : Geodatanormen - Grunnlagsnett
 Normklasse : II : Tettbygd (p= 20.0 ppm k= 20.0 mm)

STATISTIKK

Antall observasjoner retning : 11
 Antall observasjoner avstand : 4
 Antall observasjoner : 15
 Antall ukjente grunnrisskoordinater : 4
 Antall ukjente orienteringselementer : 4
 Antall ukjente : 8
 Antall overbestemmelser : 7

Maksimalt antall punkt pr. sektor : 3
 Grenseavstand for nabopunkt : 10000.000
 Antall enkeltforbindelser : 20
 Antall dobbeltforbindelser : 30

B.6.5 Example, external reliability. Network B. In Norwegian:

Ytre pålitelighet, nett B

NB: Grovfeil er bestemt med konfidensintervall-metoden

UTJEVNING I GRUNNRIS

TVUNGEN UTJEVNING

NYBESTEMTE KOORDINATER MED MIDLERE FEIL [meter]

PUNKT	X	Y	H	sX	sY	sH
P	639236.596	-513.590		0.013	0.014	
Q	635014.533	87.828		0.015	0.013	

YTRE PÅLITELIGHET - KOORDINATER [meter]

COORDINAT	Observasjon....[meter/gon]	Indre pål.	Ytre pål.
X P	B 1 P	D -0.134	-0.072*
Y P	B 1 P	D -0.134	-0.088*
P P	B 1 P	D -0.134	0.114**
X Q	P 1 Q	D 0.105	-0.034
Y Q	C 1 Q	D -0.082	0.045
P Q	C 1 Q	D -0.082	0.055*

Tabellverdi=2.16 (Student-t, f=13, alfa=0.0250)

YTRE PÅLITELIGHET - RETNING, AVSTAND OG HØYDEFORSKJELL [gon / m/km]

Forbindelse Observasjon....[meter/gon] Indre pål. Ytre pål.

R	A	Q	B	Q	D	-0.060	-0.00054
D	A	Q	C	Q	D	-0.082	0.013
R	A	P	B	P	D	-0.134	-0.00057
D	A	P	B	P	D	-0.134	-0.011
R	B	Q	C	Q	D	-0.082	-0.00096
D	B	Q	B	Q	D	-0.060	-0.010
R	B	P	C	P	D	-0.089	0.00104
D	B	P	B	P	D	-0.134	-0.041 *
R	Q	A	B	Q	D	-0.060	-0.00054
D	Q	A	C	Q	D	-0.082	0.013
R	Q	B	C	Q	D	-0.082	-0.00096
D	Q	B	B	Q	D	-0.060	-0.010
R	Q	C	B	Q	D	-0.060	0.00052
D	Q	C	C	Q	D	-0.082	-0.013
R	Q	P	B	P	D	-0.134	-0.00127
D	Q	P	P	Q	D	0.105	0.016
R	C	Q	B	Q	D	-0.060	0.00052
D	C	Q	C	Q	D	-0.082	-0.013
R	C	P	B	P	D	-0.134	-0.00129
D	C	P	C	P	D	-0.089	-0.009
R	P	A	B	P	D	-0.134	-0.00057
D	P	A	B	P	D	-0.134	-0.011
R	P	B	C	P	D	-0.089	0.00104
D	P	B	B	P	D	-0.134	-0.041 *
R	P	Q	B	P	D	-0.134	-0.00127
D	P	Q	P	Q	D	0.105	0.016
R	P	C	B	P	D	-0.134	-0.00129
D	P	C	C	P	D	-0.089	-0.009

Tabellverdi=2.16 (Student-t, f=13, alfa=0.0250)

YTRE PÅLITELIGHET - MÅLESTOKKDIFFERANSER [ppm]

Punkt	Forb1	Forb2	Normkrav	YP-M.diff.
	Observasjon....[meter/gon]		Indre pål.	

A	Q	C	20.88	-12.54
	D C	1 Q	-0.082	
B	Q	P	23.57	-44.63 *
	D B	1 P	-0.134	
Q	A	C	23.01	-25.31 *
	D C	1 Q	-0.082	
C	Q	P	21.25	14.73
	D C	1 Q	-0.082	
P	B	C	23.38	47.85 **
	D B	1 P	-0.134	

Tabellverdi=2.16 (Student-t, f=13, alfa=0.0250)

Normparametre:

Normtype : Geodatanormen - Grunnlagsnett
 Normklasse : II : Tettbygd (p= 20.0 ppm k= 20.0 mm)

YTRE PÅLITELIGHET - VINKELDEFORMASJONER [ppm]

Punkt	Forb1	Forb2	Normkrav	YP-V.def.
	Observasjon....[meter/gon]		Indre pål.	

A	Q	P	20.90	-11.60
	D B	1 P	-0.134	
B	Q	P	23.57	-18.92
	D P	1 Q	0.105	
Q	B	P	21.61	-28.01 *
	D B	1 P	-0.134	
C	B	P	20.64	-20.23
	D B	1 P	-0.134	
P	B	Q	22.55	-17.95
	D B	1 P	-0.134	

Tabellverdi=2.16 (Student-t, f=13, alfa=0.0250)

Normparametre:

Normtype : Geodatanormen - Grunnlagsnett
 Normklasse : II : Tettbygd (p= 20.0 ppm k= 20.0 mm)

STATISTIKK

Antall observasjoner retning	:	17
Antall observasjoner avstand	:	6
Antall observasjoner	:	23
Antall ukjente grunnrisskoordinater	:	4
Antall ukjente orienteringselementer	:	5
Antall ukjente	:	9
Antall overbestemmelser	:	14
Maksimalt antall punkt pr. sektor	:	3
Grenseavstand for nabopunkt	:	10000.000
Antall enkeltforbindelser	:	20
Antall dobbeltforbindelser	:	30